In The Name of God



Sharif University of Technology
Electrical Engineering Department

Electronics 3

Simulation Project 1

Professor: Dr. Mohammad Fakharzadeh

Name: Sanaz Mazrae

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Problem 1

Draw the circuit in *ADS* using the specified parameters for each transistor. Then, perform the following tasks:

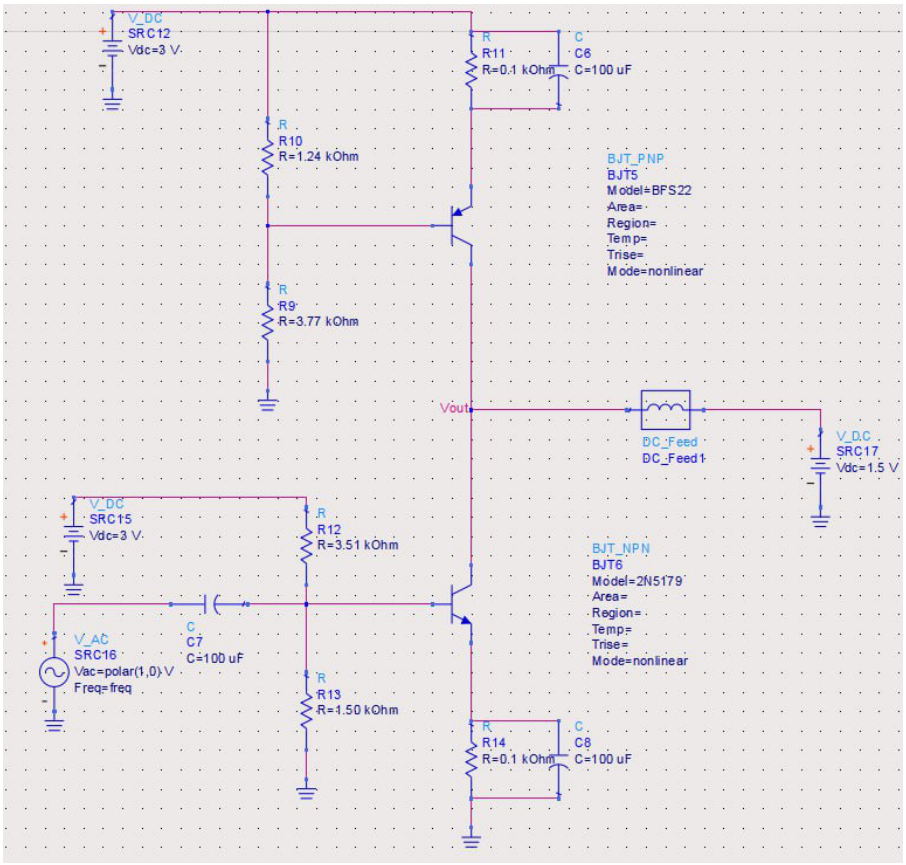


Figure 1: Common Emitter Amplifier with active load

a. Determine the operating points of the devices and report the results. Additionally, compute the small-signal parameters and the values of parasitic capacitors using the given equations, based on the I_C values observed from the operating points.

$$C_\pi \approx g_m.TF + 2.C_{JE}$$

$$C_\mu \approx \frac{1}{r_o}.TR + C_{JC}$$

$$C_{j,BE} = \frac{C_{j,BE0}}{(1+|\frac{V_{BE}}{\phi_{i,BE}}|)^m}, \quad \text{with m} = \text{MJE}$$

$$C_{j,BC} = \frac{C_{j,BC0}}{(1+|\frac{V_{BC}}{\phi_{i,BC}}|)^m}, \quad \text{with m} = \text{MJC}$$

First, we create the models of the transistors that we are going to use with *BJT_Model* box as you see in 2.

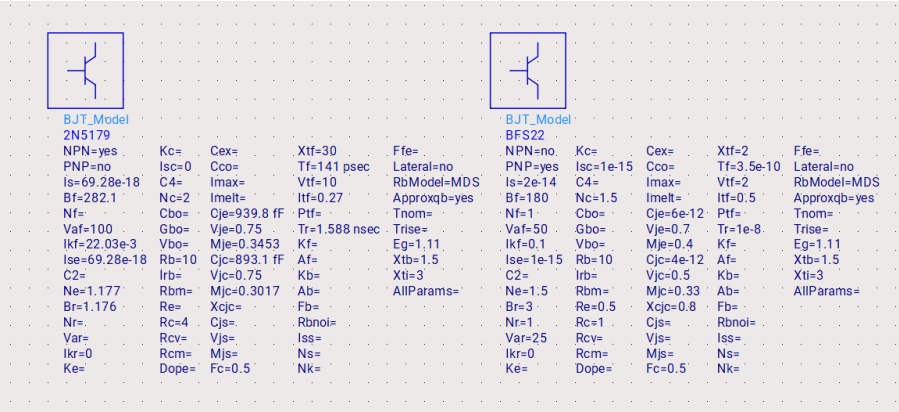


Figure 2: Transistor models

I used *DC simulation* to obtain the operating points of the transistors. Two Current probes are used in order to measure I_C of the transistors.

You can see the circuit and the results from DC simulation in 3 and 4 respectively.

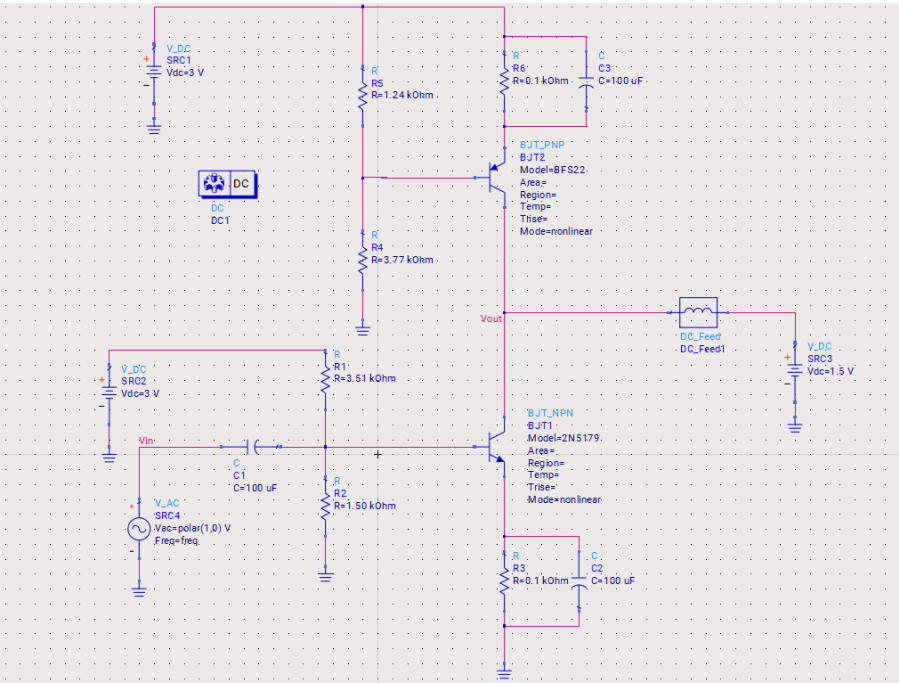


Figure 3: DC simulation circuit

index	Ic	Vbc	Vbe	Vce
1	1.013 m	-617.2 m	780.0 m	1.397
2	-1.020 m	762.9 m	-634.5 m	-1.397

Figure 4: DC simulation results

Now that we have the I_C of the transistors, we calculate the small-signal parameters:

$$g_{m_1} = \frac{I_{C_1}}{V_T} = \frac{1.013^{mA}}{26^{mV}} \approx 39^{m\mho}, \quad g_{m_2} = \frac{I_{C_2}}{V_T} = \frac{1.020^{mA}}{26^{mV}} \approx 39.2^{m\mho}$$

$$r_{\pi_1} = \frac{\beta_1}{g_{m_1}} = \frac{282.1}{39^{m\mho}} \approx 7.23^{k\Omega}, \quad r_{\pi_2} = \frac{\beta_2}{g_{m_2}} = \frac{180}{39.2^{m\mho}} \approx 4.59^{k\Omega}$$

$$r_{o_1} = \frac{V_{A_1}}{I_{C_1}} = \frac{100^V}{1.013^{mA}} \approx 98.7^{k\Omega}, \quad r_{o_2} = \frac{V_{A_2}}{I_{C_2}} = \frac{50^V}{1.020^{mA}} \approx 49^{k\Omega}$$

	V_{CE}	I_C	g_m	r_π	r_o
Q_1	1.397^V	1.013^{mA}	$39^{m\mho}$	$7.23^{k\Omega}$	$98.7^{k\Omega}$
Q_2	-1.397^V	1.020^{mA}	$39.2^{m\mho}$	$4.59^{k\Omega}$	$49^{k\Omega}$

Table 1: Bias points and small-signal parameters

Now is time to compute the values of the parasitic capacitors:

$$C_{\pi_1} = g_{m_1}.TF_1 + 2.\frac{C_{j,BE0}}{(1+\frac{V_{BE_1}}{\phi_{i,BE}})^m} = 39^{m\mho}.141^{psec} + 2.\frac{939.8^{fF}}{(1+\frac{780^{mV}}{0.75^V})^{0.3453}} \approx 6.97^{pF}$$

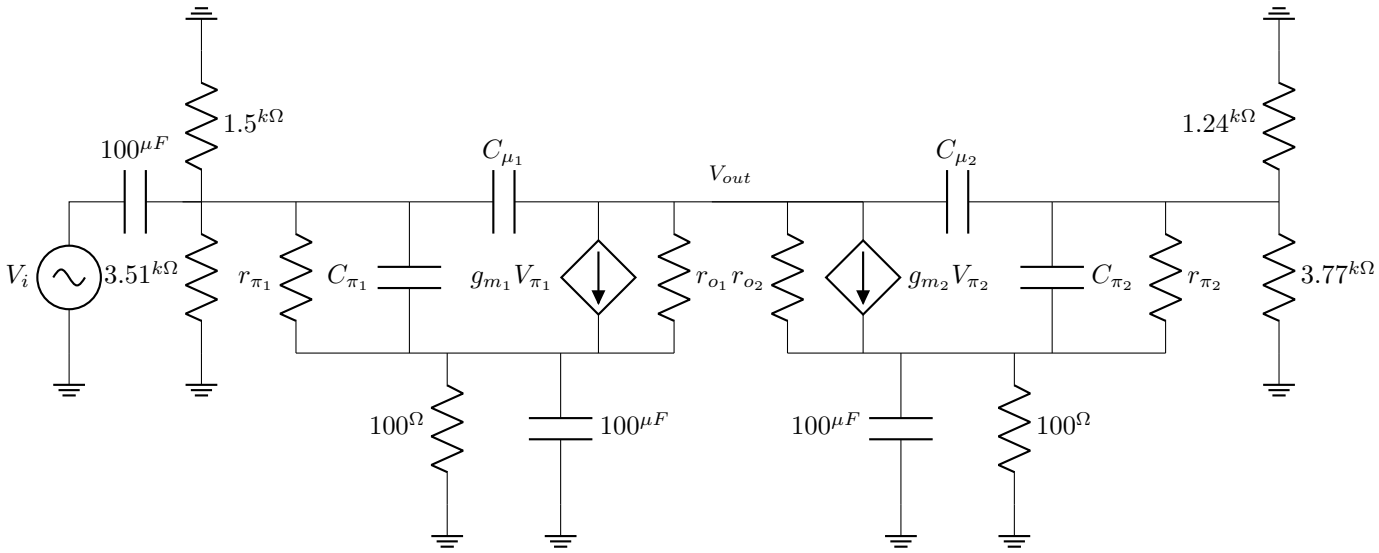
$$C_{\pi_2} = g_{m_2}.TF_2 + 2.\frac{C_{j,BE0}}{(1+\frac{V_{BE_2}}{\phi_{i,BE}})^m} = 39.2^{m\mho}.350^{psec} + 2.\frac{6^{pF}}{(1+\frac{634^{mV}}{0.7^V})^{0.4}} \approx 23^{pF}$$

$$C_{\mu_1} \approx \frac{1}{r_{o_1}}.TR_1 + \frac{C_{j,BC0}}{(1+\frac{V_{BC_1}}{\phi_{i,BC}})^m} = \frac{1.588^{nsec}}{98.7^{k\Omega}} + \frac{893.1^{fF}}{(1+\frac{617.2^{mV}}{0.75})^{0.3017}} \approx 0.761^{pF}$$

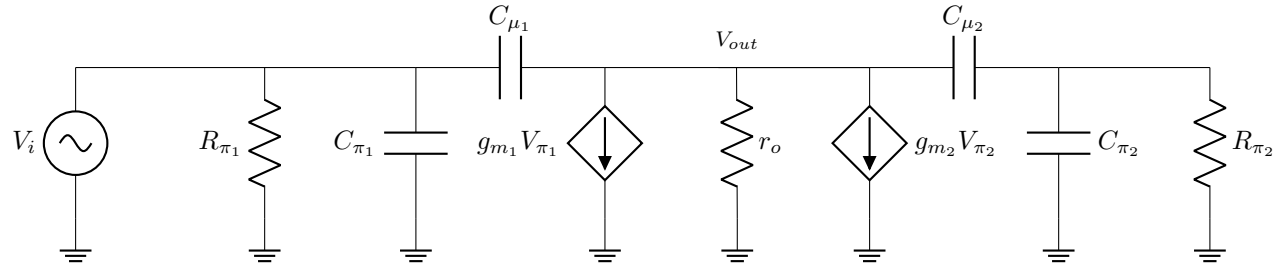
$$C_{\mu_2} \approx \frac{1}{r_{o_2}}.TR_2 + \frac{C_{j,BC0}}{(1+\frac{V_{BC_2}}{\phi_{i,BC}})^m} = \frac{10^{nsec}}{49^{k\Omega}} + \frac{4^{pF}}{(1+\frac{763^{mV}}{0.5})^{0.33}} \approx 3.15^{pF}$$

b. Derive the theoretical parametric transfer function of the circuit. Then, using the values obtained in the previous step, plot the **Bode diagram** of the circuit in *MATLAB* (both amplitude and phase) and report f_{3dB} and f_u for this circuit.

Lets draw the draw the small-signal equivalent circuit using the h-π model.



Considering the $100^{\mu F}$ capacitors as short circuit the circuit simplifies to:



where: $r_o = r_{o_1}||r_{o_2}$, $R_{\pi_1} = r_{\pi_1}||3.51^{k\Omega}||1.5^{k\Omega}$ and $R_{\pi_2} = r_{\pi_2}||1.24^{k\Omega}||3.77^{k\Omega}$

Now lets derive the parametric transfer function of the circuit above.

KCL @ V_{out}: $\frac{V_{out}}{r_o} + g_{m_1}V_i + g_{m_2}V_{\pi_2} + \frac{V_{out}-V_i}{\frac{1}{C_{\mu_1}s}} + \frac{V_{out}-V_{\pi_2}}{\frac{1}{C_{\mu_2}s}} = 0$

KCL @ V_{b_2}: $\frac{V_{\pi_2}-V_{out}}{\frac{1}{C_{\mu_2}s}} + \frac{V_{\pi_2}}{R_{\pi_2}} + \frac{V_{\pi_2}}{\frac{1}{C_{\pi_2}s}} = 0 \Longrightarrow V_{out} = \frac{(C_{\mu_2}+C_{\pi_2})s+\frac{1}{R_{\pi_2}}}{C_{\mu_2}s}V_{\pi_2}$

$\longrightarrow V_{out} = \frac{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}{R_{\pi_2}C_{\mu_2}s}V_{\pi_2} \Longrightarrow V_{\pi_2} = \frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}V_{out}$

$\longrightarrow \frac{V_{out}}{r_o} + g_{m_1}V_i + g_{m_2}\frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}V_{out} + C_{\mu_1}s(V_{out}-V_i) + \dots$

$C_{\mu_2}s(1 - \frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1})V_{out} = 0 \rightarrow$ assume $K = \frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}$

$(\frac{1}{r_o} + g_{m_2}K + C_{\mu_1}s + C_{\mu_2}s(1-K))V_{out} = (C_{\mu_1}s - g_{m_1})V_i \Longrightarrow \dots$

$$A_v(s) = \frac{V_{out}}{V_i}(s) = \frac{C_{\mu_1}s - g_{m_1}}{\frac{1}{r_o} + g_{m_2}K + C_{\mu_1}s + C_{\mu_2}s(1-K)}$$

$$= \frac{C_{\mu_1}s - g_{m_1}}{\frac{1}{r_o} + g_{m_2}\frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1} + C_{\mu_1}s + C_{\mu_2}s\frac{R_{\pi_2}C_{\pi_2}s+1}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}}$$

$$= \frac{r_o(C_{\mu_1}s - g_{m_1})}{1 + r_o g_{m_2}\frac{R_{\pi_2}C_{\mu_2}s}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1} + r_o C_{\mu_1}s + r_o C_{\mu_2}s\frac{R_{\pi_2}C_{\pi_2}s+1}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1}}$$

$$= \frac{r_o(C_{\mu_1}s - g_{m_1})(R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1)}{R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1 + r_o g_{m_2}R_{\pi_2}C_{\mu_2}s + r_o C_{\mu_1}s(R_{\pi_2}(C_{\mu_2}+C_{\pi_2})s+1) + r_o C_{\mu_2}s(R_{\pi_2}C_{\pi_2}s+1)}$$

$$= \frac{r_o R_{\pi_2}C_{\mu_1}(C_{\mu_2}+C_{\pi_2})s^2 + r_o(C_{\mu_1} - g_{m_1}R_{\pi_2}(C_{\mu_2}+C_{\pi_2}))s - r_o g_{m_1}}{r_o R_{\pi_2}(C_{\mu_1}C_{\mu_2}+C_{\mu_1}C_{\pi_2}+C_{\mu_2}C_{\pi_2})s^2 + [R_{\pi_2}(C_{\mu_2}+C_{\pi_2}) + r_o g_{m_2}R_{\pi_2}C_{\mu_2} + r_o C_{\mu_1} + r_o C_{\mu_2}]s + 1}$$

Now that we derived the transfer function, we put the numerical values of the parameters in the function and plot its magnitude and phase in *MATLAB* using *bode* function. Here are the results:

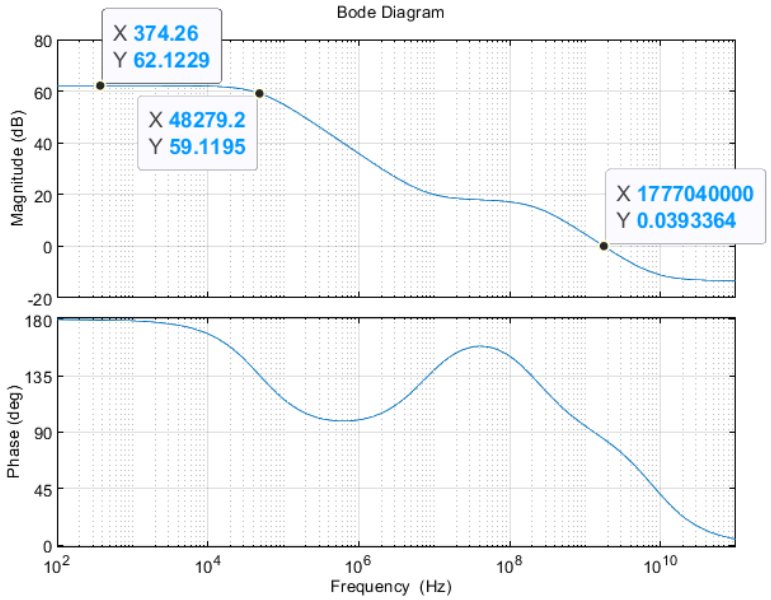


Figure 5: Bode diagram of the derived transfer function in *MATLAB*

As we can see from the diagram $f_{3dB} \approx 48^{kHz}$ and $f_u \approx 1.77^{GHz}$.

c. Perform an AC simulation in *ADS* and plot the frequency response of the circuit.

Now is time to use *AC simulation* in order to plot the output voltage of the circuit which would also give us the transfer function diagram of the circuit because the amplitude of the input voltage is 1. I set the frequency range to 100^{Hz}-100^{GHz} from the *AC simulation* box as needed.

You can see the resulting *Bode diagram* in 7.

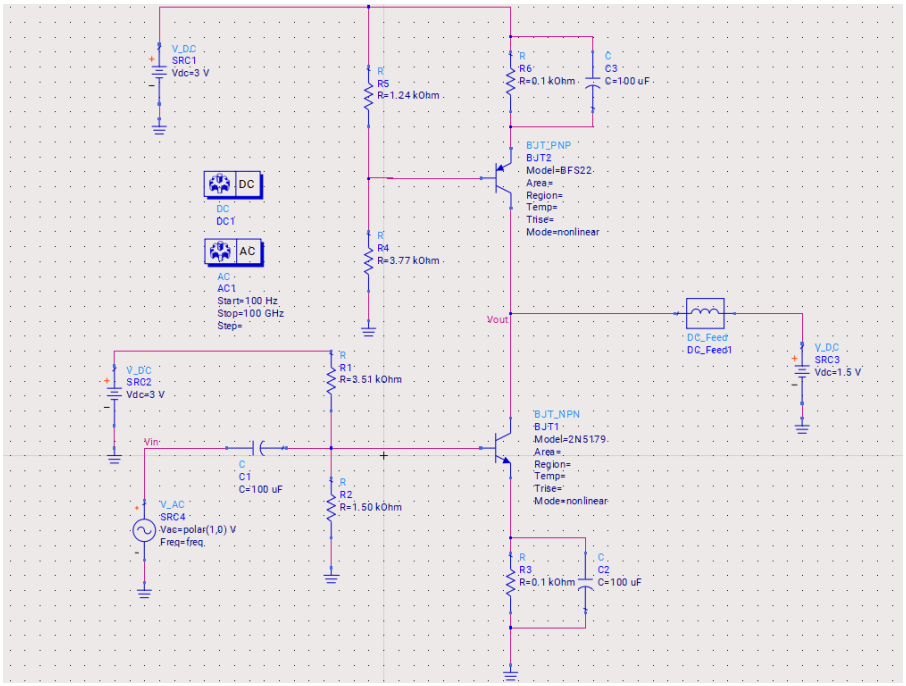


Figure 6: AC simulation circuit

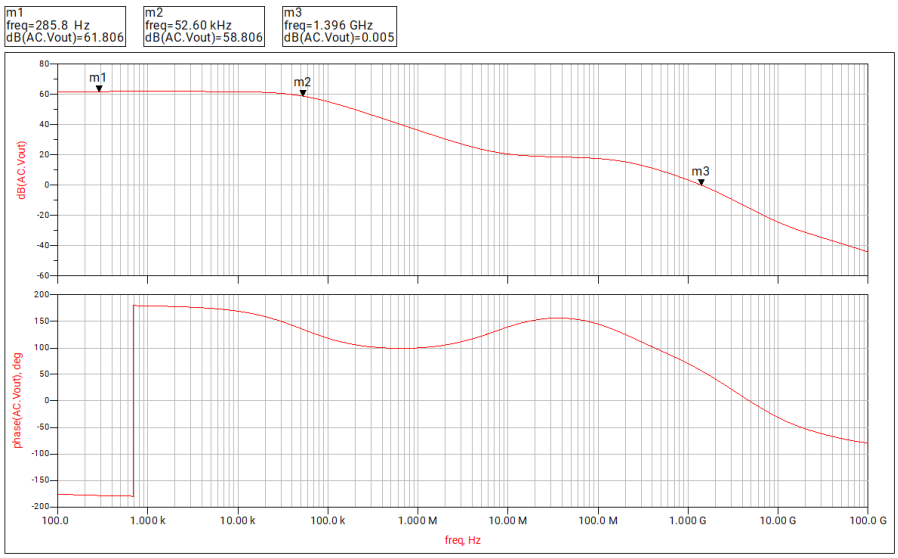


Figure 7: Bode diagram of the transfer function of the circuit in *ADS*

Results from simulation: $f_{3dB} \approx 53^{kHz}$ and $f_u \approx 1.4^{GHz}$

d. Compare the transfer function obtained from *MATLAB* in part *b* with the frequency response simulated in *ADS* in step *c* and explain the differences between the theoretical and simulated results.

The general form of the frequency responses obtained from theory and simulation is nearly the same but as we expect the results have some differences. An important reason for this is that we ignored the impact of the $100^{\mu F}$ capacitors on the frequency response. Another reason is the approximations we used in theory.

e. Place a capacitor at the output node (before the DC feed, with one end of the capacitor connected to ground). Draw the **Bode diagram** for each capacitor value and analyze its effects for the following values: $C = 1^{pF}, 10^{pF}, 100^{pF}, 1^{nF}, 10^{nF}$.

For sweeping the capacitor, i used a *Variable Equation* box and a *Parameter Sweep* box to define a variable and sweep it through the given range. Then i passed my variable named *Cap* to the capacitor and plotted the frequency response of the circuit.

You can see the circuit and the results in 8 and 9 respectively.

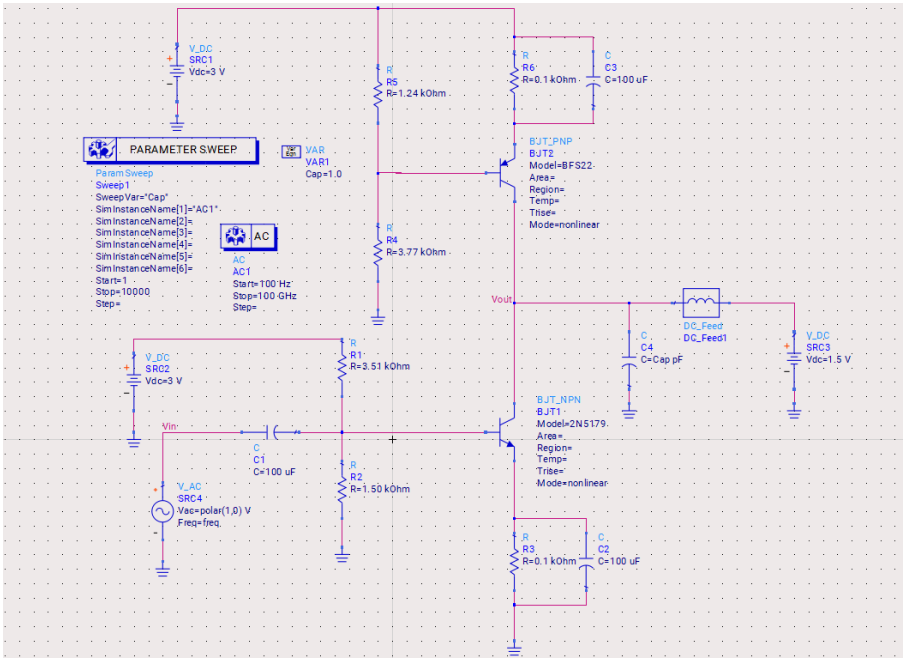


Figure 8: The circuit of part *e*

Comparing the Bode diagrams, we can see that increase in the value of the capacitor leads to a smaller output pole which gradually makes the output pole the dominant pole of the circuit determining the bandwidth. So increasing the value of the capacitor results in less bandwidth and the

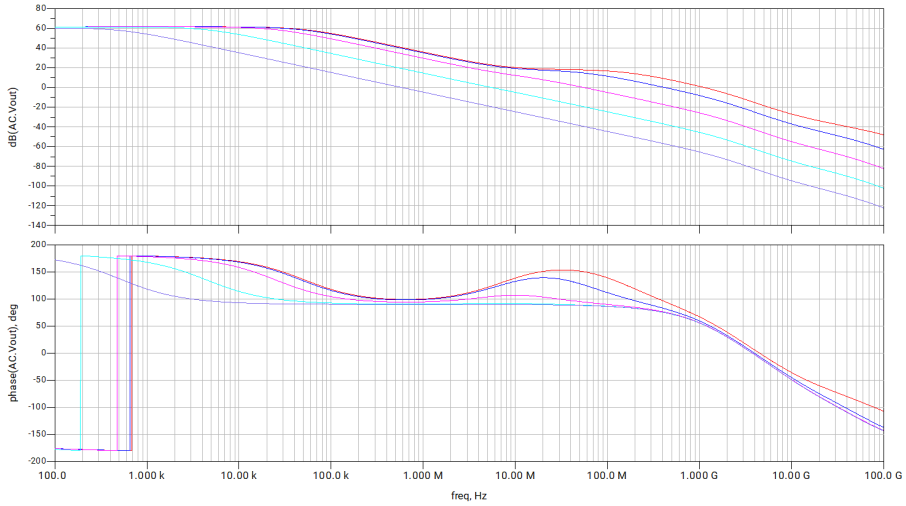


Figure 9: Bode diagram: output capacitor swept from 1^{pF} to 10^{nF}

output pole becomes the dominant pole.(This capacitor operates similar to the parallel connection of Collector-Substrate capacitors(C_{CS}) of the transistors which we neglected in this problem.)

f. By sweeping the capacitor value as done in the previous section, can you determine the frequency at which the pole of the output node would exist in the absence of the capacitor? If your answer is yes, explain your method.

Similar to the process in the previous part, I put a capacitor in the output node and then swept the value of the capacitor(just like what we did in part *e*).

The results show that a pole is moving from high frequencies to low frequencies while the value of the capacitor increases. Therefore we can associate that pole with the output node because other parameters in the circuit are stable and the only variable is the output capacitor.

Problem 2

Draw the circuit in *ADS* using the given parameters. Then, perform the following tasks:

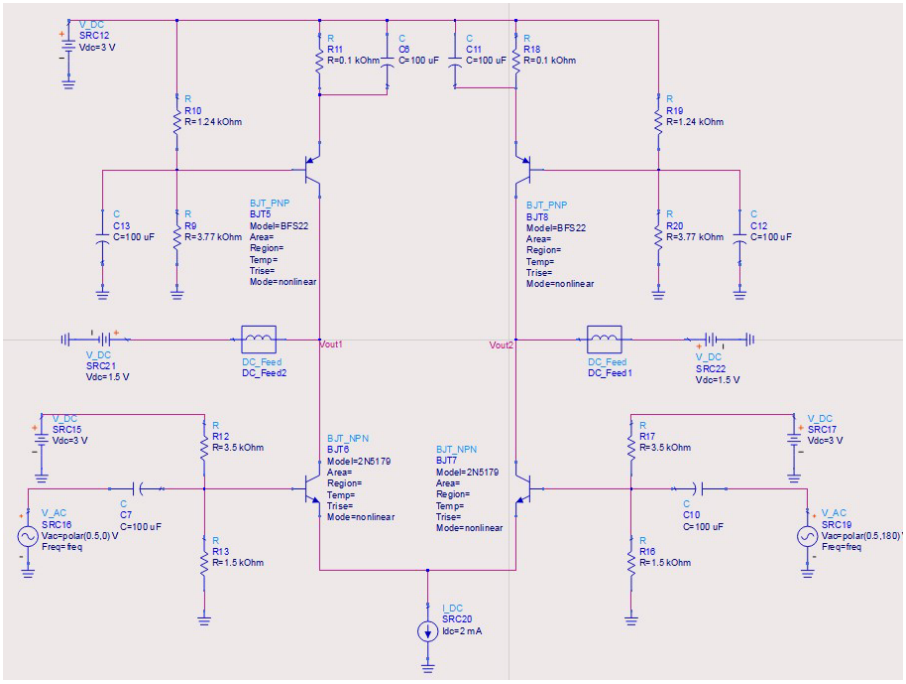


Figure 10: Differential amplifier with active load

a. Determine the operating points of the devices and report the results. Additionally, compute the small-signal parameters and the values of parasitic capacitors using the given equations, based on the I_C values observed from the operating points.

$$C_\pi \approx g_m.TF + 2.C_{JE}$$

$$C_\mu \approx \frac{1}{r_o}.TR + C_{JC}$$

$$C_{j,BE} = \frac{C_{j,BE0}}{(1+|\frac{V_{BE}}{\phi_{i,BE}}|)^m}, \quad \text{with m} = \text{MJE}$$

$$C_{j,BC} = \frac{C_{j,BC0}}{(1+|\frac{V_{BC}}{\phi_{i,BC}}|)^m}, \quad \text{with m} = \text{MJC}$$

Because of the 2^{mA} Current Source and the symmetry of the circuit, all the transistors must have $I_C \approx 1^{mA}$. So the theoretical operating points and small-signal parameters are as follows:

	V_{CE}	I_C	g_m	r_π	r_o
Q_1	–	1^{mA}	$38.5^{m\mho}$	$7.06^{k\Omega}$	$100^{k\Omega}$
Q_2	-1.4^V	1^{mA}	$38.5^{m\mho}$	$4.5^{k\Omega}$	$50^{k\Omega}$
Q_3	–	1^{mA}	$38.5^{m\mho}$	$7.06^{k\Omega}$	$100^{k\Omega}$
Q_4	-1.4^V	1^{mA}	$38.5^{m\mho}$	$4.5^{k\Omega}$	$50^{k\Omega}$

Table 2: Theoretical bias points and small-signal parameters

Now we use DC simulation to obtain the operating points by simulation. The circuit and the results are attached in figures 11 and 12.

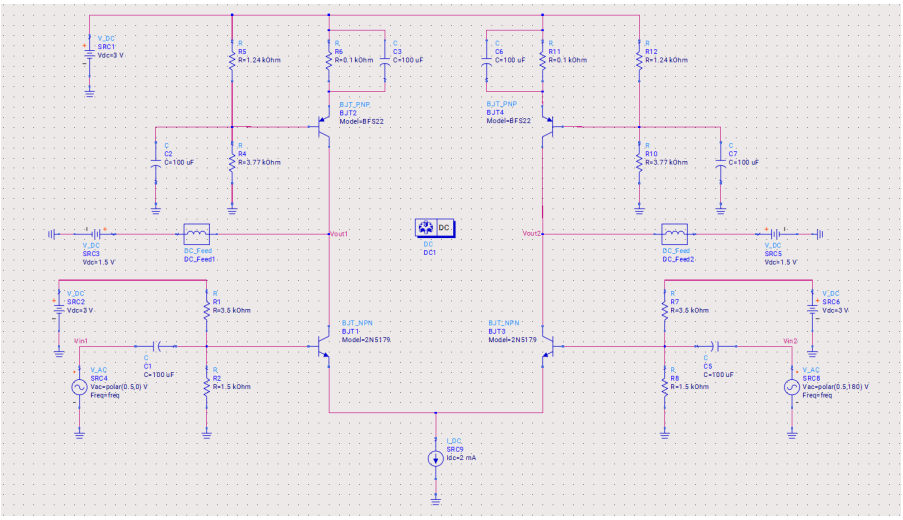


Figure 11: DC simulation circuit

index	Ic	Vbc	Vbe	Vce
1	985.7 u	-615.0 m	779.2 m	1.394
2	-1.020 m	762.9 m	-634.5 m	-1.397
3	985.7 u	-615.0 m	779.2 m	1.394
4	-1.020 m	762.9 m	-634.5 m	-1.397

Figure 12: DC simulation results

The operating points and small-signal parameters obtained from simulation:

	V_{CE}	I_C	g_m	r_π	r_o
Q_1	1.394^V	$985.7^{\mu A}$	$37.9^{m\mho}$	$7.44^{k\Omega}$	$101.5^{k\Omega}$
Q_2	-1.397^V	1.020^{mA}	$39.2^{m\mho}$	$4.59^{k\Omega}$	$49^{k\Omega}$
Q_3	1.394^V	$985.7^{\mu A}$	$37.9^{m\mho}$	$7.44^{k\Omega}$	$101.5^{k\Omega}$
Q_4	-1.397^V	1.020^{mA}	$39.2^{m\mho}$	$4.59^{k\Omega}$	$49^{k\Omega}$

Table 3: Simulation bias points and small-signal parameters

Now lets compute the values of the parasitic capacitors.

$$C_{\pi_1} = g_{m_1}.TF_1 + 2.\frac{C_{j,BE0}}{(1+\frac{V_{BE_1}}{\phi_{i,BE}})^m} = 37.9^{m\mho}.141^{psec} + 2.\frac{939.8^{fF}}{(1+\frac{779.3^{mV}}{0.75^V})^{0.3453}} \approx 6.81^{pF}$$

$$C_{\pi_2} = g_{m_2}.TF_2 + 2.\frac{C_{j,BE0}}{(1+\frac{V_{BE_2}}{\phi_{i,BE}})^m} = 39.2^{m\mho}.350^{psec} + 2.\frac{6^{pF}}{(1+\frac{634^{mV}}{0.7^V})^{0.4}} \approx 23^{pF}$$

$$C_{\mu_1} \approx \frac{1}{r_{o_1}}.TR_1 + \frac{C_{j,BC0}}{(1+\frac{V_{BC_1}}{\phi_{i,BC}})^m} = \frac{1.588^{nsec}}{101.5^{k\Omega}} + \frac{893.1^{fF}}{(1+\frac{615^{mV}}{0.75})^{0.3017}} \approx 0.761^{pF}$$

$$C_{\mu_2} \approx \frac{1}{r_{o_2}}.TR_2 + \frac{C_{j,BC0}}{(1+\frac{V_{BC_2}}{\phi_{i,BC}})^m} = \frac{10^{nsec}}{49^{k\Omega}} + \frac{4^{pF}}{(1+\frac{763^{mV}}{0.5})^{0.33}} \approx 3.15^{pF}$$

Because the circuit is symmetric:

$$C_{\pi_1} = C_{\pi_3} \quad , \quad C_{\mu_1} = C_{\mu_3} \qquad \text{and} \qquad C_{\pi_2} = C_{\pi_4} \quad , \quad C_{\mu_2} = C_{\mu_4}$$

b. Determine the **differential-mode** transfer function without any simulation or calculations, just based on the question 1.(considering an **ideal current source**)

Because of the $100^{\mu F}$ capacitor that goes from the base of Q_1 and Q_2 to ground, the differential transfer function of the circuit would be similar to that one in question 1 with $R_{\pi_2} = 0$.

$A_{v_d}(s) = \frac{V_{out1} - V_{out2}}{V_{id}} = \frac{r_o(C_{\mu_1}s - g_{m_1})}{r_o(C_{\mu_1}+C_{\mu_2})s+1}$ in which: $r_o = r_{o_1}||r_{o_2}$

Below you can observe the theoretical and simulation Bode diagrams obtained from *MATLAB* and *ADS*.

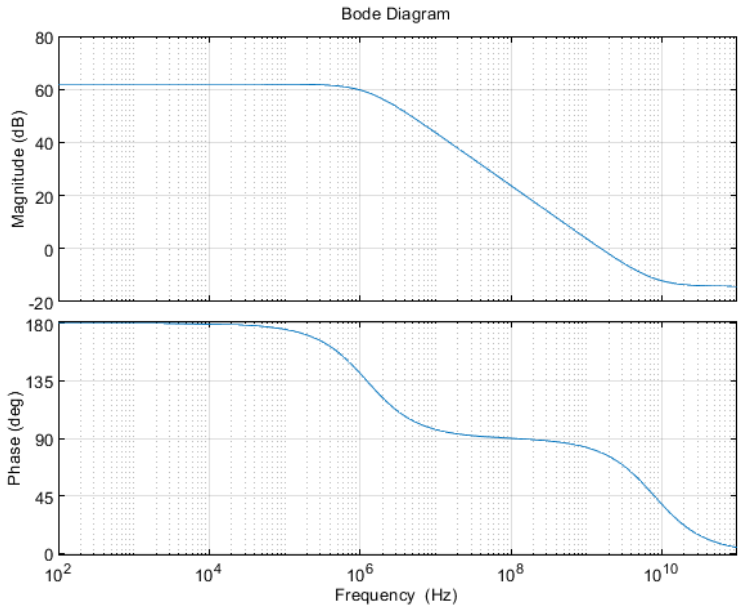


Figure 13: Bode diagram of the derived differential-mode transfer function in *MATLAB*

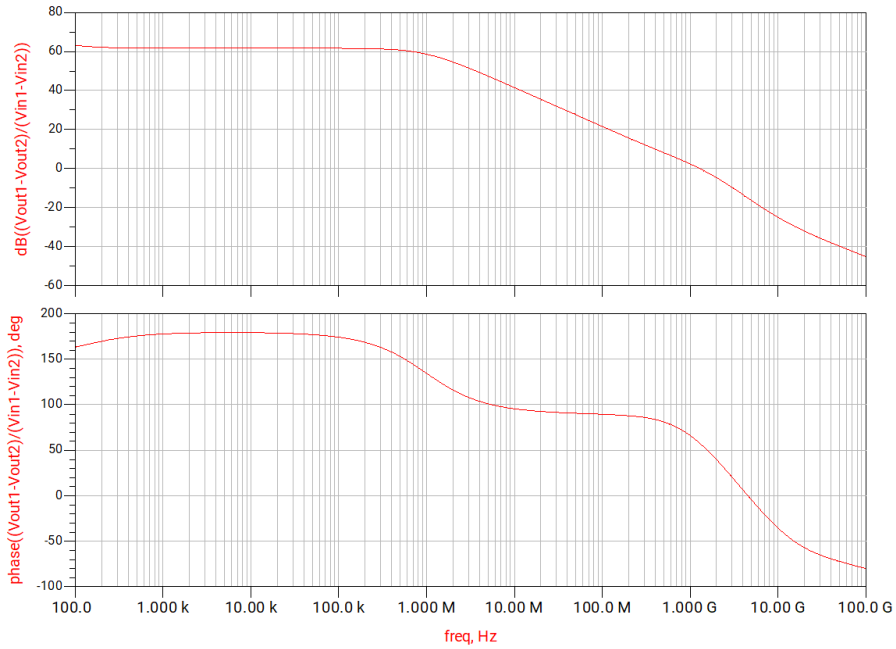


Figure 14: Bode diagram of the differential-mode transfer function of the circuit in *ADS*

c. Determine the **common-mode frequency response** and verify it using *ADS*.(considering an **ideal current source**)

When the Current Source is ideal its output impedance is ∞. Knowing this, lets do the calculations.

single-ended: $A_{v_{CM}} = \frac{V_{out1}}{V_{ic}} = \frac{r_o || \frac{1}{C_{\mu_2} s}}{r_o || \frac{1}{C_{\mu_2} s} + \frac{1}{C_{\mu_1} s}} = \frac{r_o C_{\mu_1} s}{r_o (C_{\mu_1} + C_{\mu_2}) s + 1}$

double-ended: $A_{v_{CM}} = \frac{V_{out1} - V_{out2}}{V_{ic}} = 0$

Now lets plot the Bode Diagrams in *MATLAB* and *ADS*.

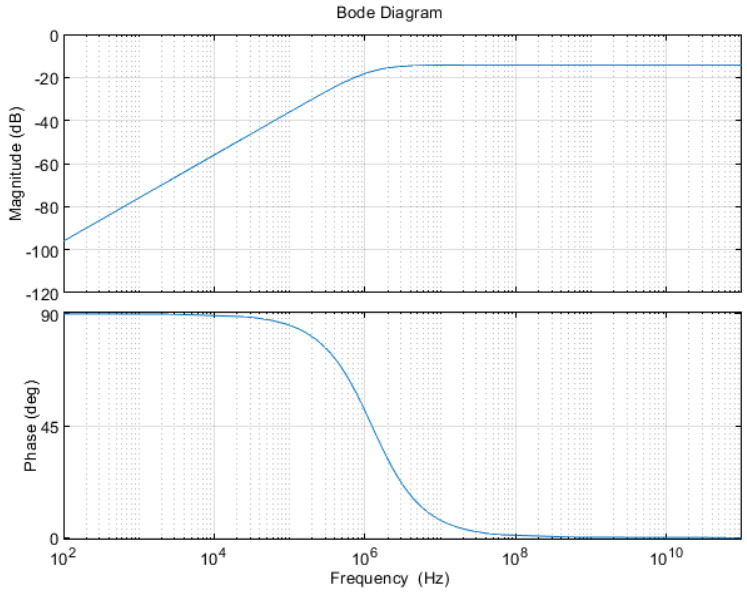


Figure 15: Bode diagram of the derived common-mode transfer function in *MATLAB*

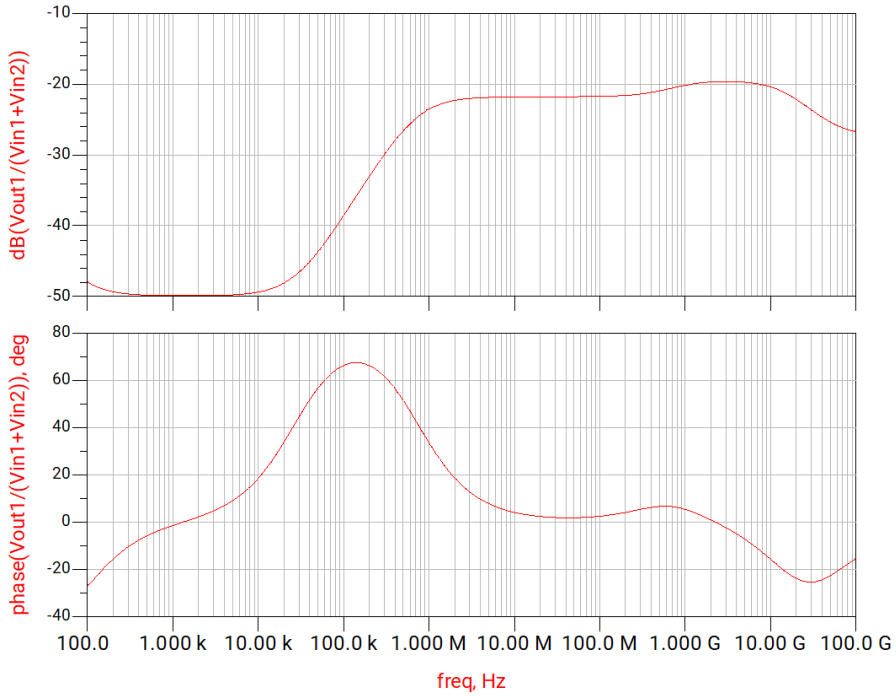


Figure 16: Bode diagram of the common-mode transfer function of the circuit in *ADS* (single-ended)

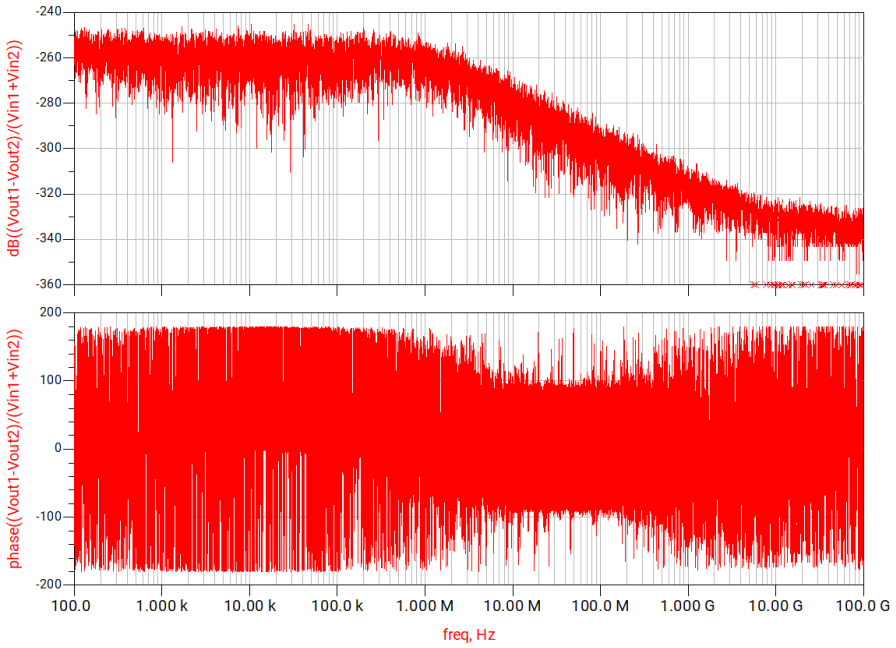


Figure 17: Bode diagram of the common-mode transfer function of the circuit in *ADS* (double-ended)

As we expect the double-ended common-mode gain is 0. The single-ended common-mode transfer function from theory and simulation are generally the same but we can see some notable differences which mostly are because of the approximations we used to derive the theoretical transfer function.

d. Repeat parts *b* and *c*, this time considering a **non-ideal current source**.

In this part we need to model a non-ideal current source. I used a $30^{k\Omega}$ resistor in series and a 1^{pF} capacitor in parallel followed by a DC-Block to preserve the operating points.

The constructed circuit of this part is demonstrated in figure 18.

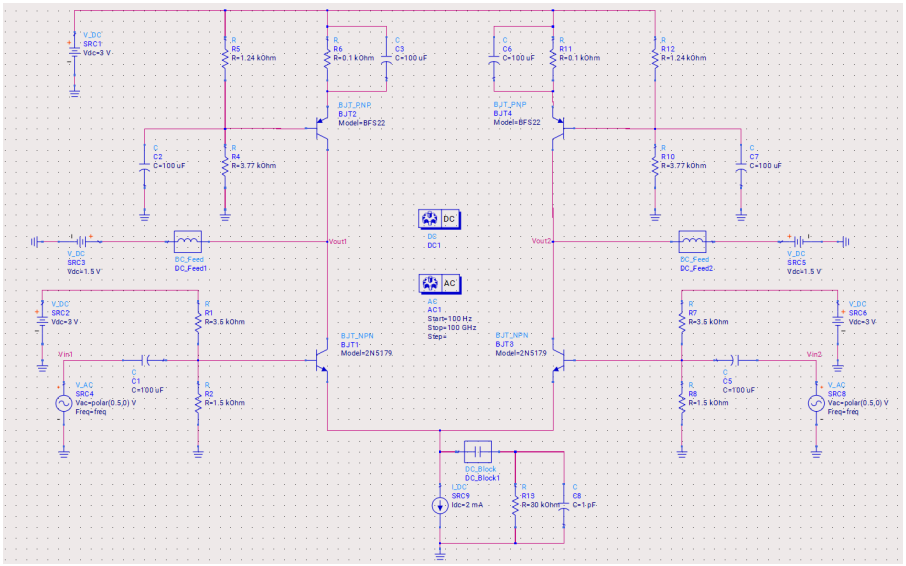
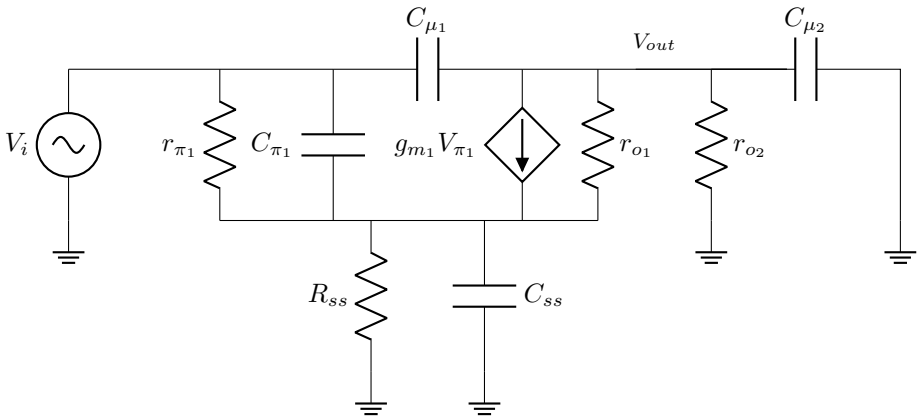


Figure 18: The constructed circuit of part d

Now lets do the calculations and derive the common-mode transfer function. The differential-mode transfer function will be the same as the one derived in part *b*.

The h-π model of the half-circuit is as follows:



$Z_{ss} = R_{ss}||\frac{1}{C_{ss}s}, Z_{\pi_1} = r_{\pi_1}||\frac{1}{C_{\pi_1}s}, Z_2 = r_{o_2}||\frac{1}{C_{\mu_2}s}, \quad R_{ss} = 1^{M\Omega}, C_{ss} = 100^{\mu F}$

KCL @ e_1 node: $\frac{V_e}{Z_{ss}} + \frac{V_e - V_i}{Z_{\pi_1}} + \frac{V_e - V_{out}}{r_{o_1}} - g_{m_1}(V_i - V_e) = 0$

KCL @ output node: $\frac{V_{out}}{Z_2} + \frac{V_{out} - V_e}{r_{o_1}} + g_{m_1}(V_i - V_e) + \frac{V_{out} - V_i}{\frac{1}{C_{\mu_1}s}} = 0$

$$\Rightarrow (\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s)V_{out} + (g_{m_1} - C_{\mu_1}s)V_i = (\frac{1}{r_{o_1}} + g_{m_1})V_e$$

$$\Rightarrow V_e = \frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{\frac{1}{r_{o_1}} + g_{m_1}} V_{out} + \frac{g_{m_1} - C_{\mu_1}s}{r_{o_1} + g_{m_1}} V_i$$

$$\Rightarrow [\frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{ss}(\frac{1}{r_{o_1}} + g_{m_1})} + \frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{\pi_1}(\frac{1}{r_{o_1}} + g_{m_1})} - \frac{1}{r_{o_1}} + g_{m_1}(\frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{\frac{1}{r_{o_1}} + g_{m_1}})]V_{out}$$

$$= [\frac{C_{\mu_1}s - g_{m_1}}{Z_{ss}(\frac{1}{r_{o_1}} + g_{m_1})} + \frac{C_{\mu_1}s - g_{m_1}}{Z_{\pi_1}(\frac{1}{r_{o_1}} + g_{m_1})} + \frac{C_{\mu_1}s - g_{m_1}}{r_{o_1}(\frac{1}{r_{o_1}} + g_{m_1})} + \frac{1}{Z_{\pi_1}} + g_{m_1} - g_{m_1}\frac{g_{m_1} - C_{\mu_1}s}{\frac{1}{r_{o_1}} + g_{m_1}}]V_i$$

$$\Rightarrow A_{v_{CM}} = \frac{V_{out}}{V_i} = \frac{\frac{C_{\mu_1}s - gm_1}{Z_{ss}(\frac{1}{r_{o_1}} + gm_1)} + \frac{C_{\mu_1}s - gm_1}{Z_{\pi_1}(\frac{1}{r_{o_1}} + gm_1)} + \frac{C_{\mu_1}s - gm_1}{r_{o_1}(\frac{1}{r_{o_1}} + gm_1)} + \frac{1}{Z_{\pi_1}} + g_{m_1} - g_{m_1}\frac{gm_1 - C_{\mu_1}s}{\frac{1}{r_{o_1}} + gm_1}}{\frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{ss}(\frac{1}{r_{o_1}} + gm_1)} + \frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{\pi_1}(\frac{1}{r_{o_1}} + gm_1)} - \frac{1}{r_{o_1}} + g_{m_1}(\frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{\frac{1}{r_{o_1}} + gm_1})}$$

$$= \frac{\frac{C_{\mu_1}s - gm_1}{Z_{ss}} + \frac{C_{\mu_1}s - gm_1}{Z_{\pi_1}} + \frac{C_{\mu_1}s - gm_1}{r_{o_1}} + \frac{\frac{1}{r_{o_1}} + gm_1}{Z_{\pi_1}} + g_{m_1}(\frac{1}{r_{o_1}} + g_{m_1}) - g_{m_1}(g_{m_1} - C_{\mu_1}s)}{\frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{ss}} + \frac{\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{Z_{\pi_1}} - \frac{\frac{1}{r_{o_1}} + gm_1}{r_{o_1}} + g_{m_1}(\frac{1}{Z_2} + \frac{1}{r_{o_1}} + C_{\mu_1}s)}$$

$$= \frac{\frac{C_{\mu_1}s - gm_1}{\frac{R_{ss}}{R_{ss}C_{ss}s+1}} + \frac{C_{\mu_1}s + \frac{1}{r_{o_1}}}{\frac{r_{\pi_1}}{r_{\pi_1}C_{\pi_1}s+1}} + \frac{C_{\mu_1}s - gm_1}{r_{o_1}} + g_{m_1}(\frac{1}{r_{o_1}} + g_{m_1}) - g_{m_1}(g_{m_1} - C_{\mu_1}s)}{\frac{\frac{1}{r_{o_2}}}{\frac{r_{o_2}C_{\mu_2}s+1}} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{\frac{R_{ss}}{R_{ss}C_{ss}s+1}} + \frac{\frac{\frac{1}{r_{o_2}}}{\frac{r_{o_2}C_{\mu_2}s+1}} + \frac{1}{r_{o_1}} + C_{\mu_1}s}{\frac{r_{\pi_1}}{r_{\pi_1}C_{\pi_1}s+1}} - \frac{\frac{1}{r_{o_1}} + gm_1}{r_{o_1}} + g_{m_1}(\frac{\frac{1}{r_{o_2}}}{\frac{r_{o_2}C_{\mu_2}s+1}} + \frac{1}{r_{o_1}} + C_{\mu_1}s)} = \dots$$

$$\frac{R_{ss}r_{o_1}r_{\pi_1}C_{\mu_1}(C_{\pi_1}+C_{ss})s^2+[C_{\mu_1}r_{o_1}(R_{ss}+r_{\pi_1})+R_{ss}r_{\pi_1}(C_{\mu_1}+C_{\pi_1})+R_{ss}r_{o_1}g_{m_1}r_{\pi_1}(C_{\mu_1}-C_{ss})]s+R_{ss}-r_{o_1}g_{m_1}r_{\pi_1}}{R_{ss}r_{o_1}r_{\pi_1}[C_{\mu_1}C_{\pi_1}+C_{\mu_2}C_{\pi_1}+C_{\mu_1}C_{ss}+C_{\mu_2}C_{ss}]s^2+[\frac{R_1}{r_{o_2}}(C_{\mu_1}+C_{\mu_2})+\frac{R_2}{r_{o_2}}(C_{ss}+C_{\pi_1})]s+\frac{R_3}{r_{o_2}}}$$

where:

$$R_1 = R_{ss}r_{o_1}r_{o_2} + R_{ss}r_{o_2}r_{\pi_1} + r_{o_1}r_{o_2}r_{\pi_1} + R_{ss}g_{m_1}r_{o_1}r_{o_2}r_{\pi_1}$$

$$R_2 = R_{ss}r_{o_1}r_{\pi_1} + R_{ss}r_{o_2}r_{\pi_1}$$

$$R_3 = R_{ss}(r_{o_1} + r_{o_2} + r_{\pi_1} + g_{m_1}r_{o_1}r_{\pi_1}) + r_{\pi_1}(r_{o_1} + r_{o_2})$$

Now we put the numerical values of the parameters and plot the Bode diagram in *MATLAB* and then also in *ADS*.

You can observe the results in the following figures.

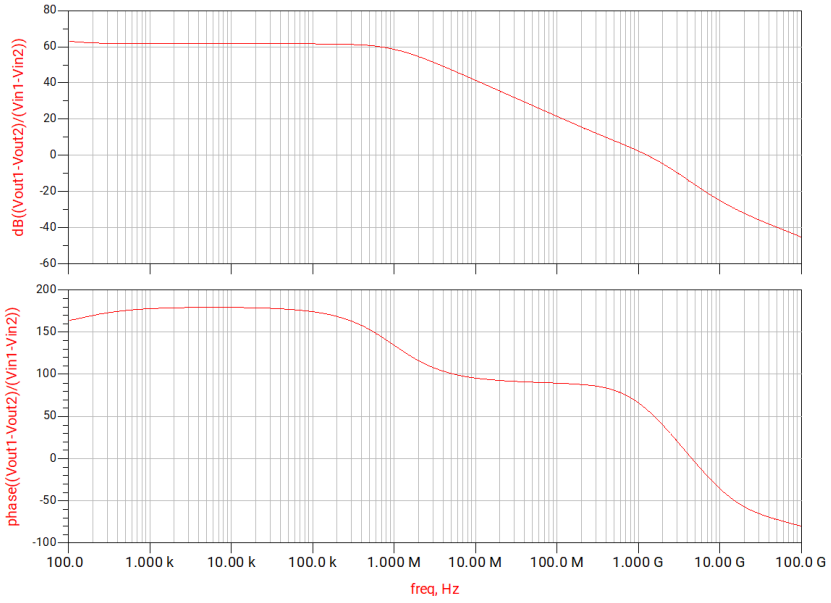


Figure 19: Bode diagram of the differential-mode transfer function with non-ideal source

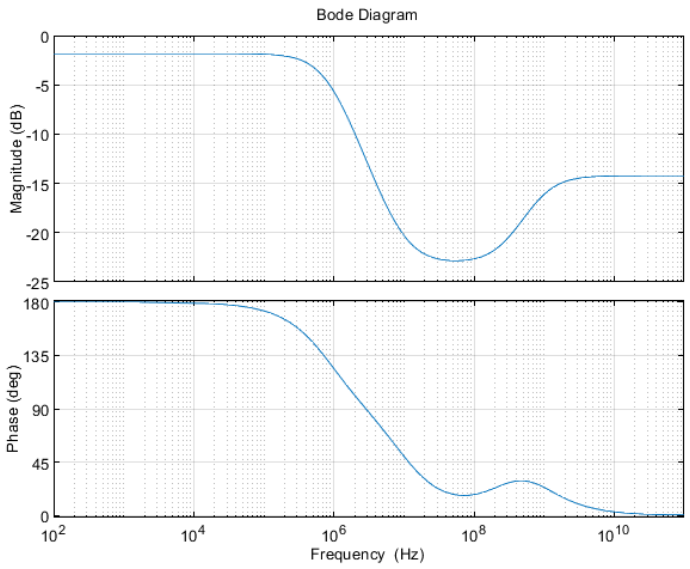


Figure 20: Bode diagram of the derived common-mode transfer function with non-ideal source (single-ended)

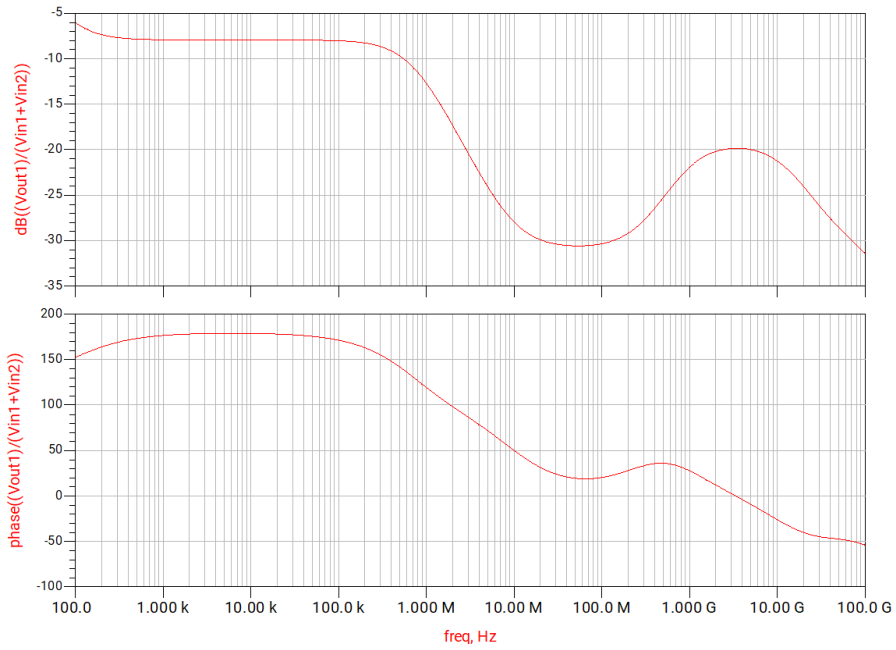


Figure 21: Bode diagram of the common-mode transfer function with non-ideal source (single-ended)

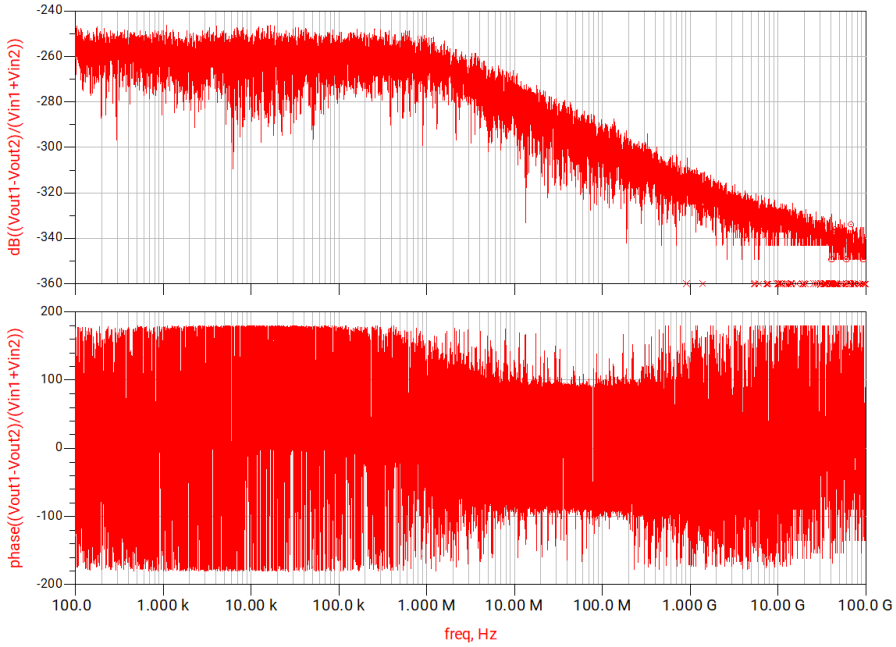


Figure 22: Bode diagram of the common-mode transfer function with non-ideal source (double-ended)

It is obvious in the results the theoretical common-mode gain is quite similar to the simulation result except for the high frequencies of 10^{GHz} and above which may be due to our negligence of other capacitors in our calculations and also the approximations of theory.

Problem 3

In this problem, we have a differential amplifier with a current mirror active load. Draw the following circuit in *ADS* with the desired parameters and perform the following tasks.

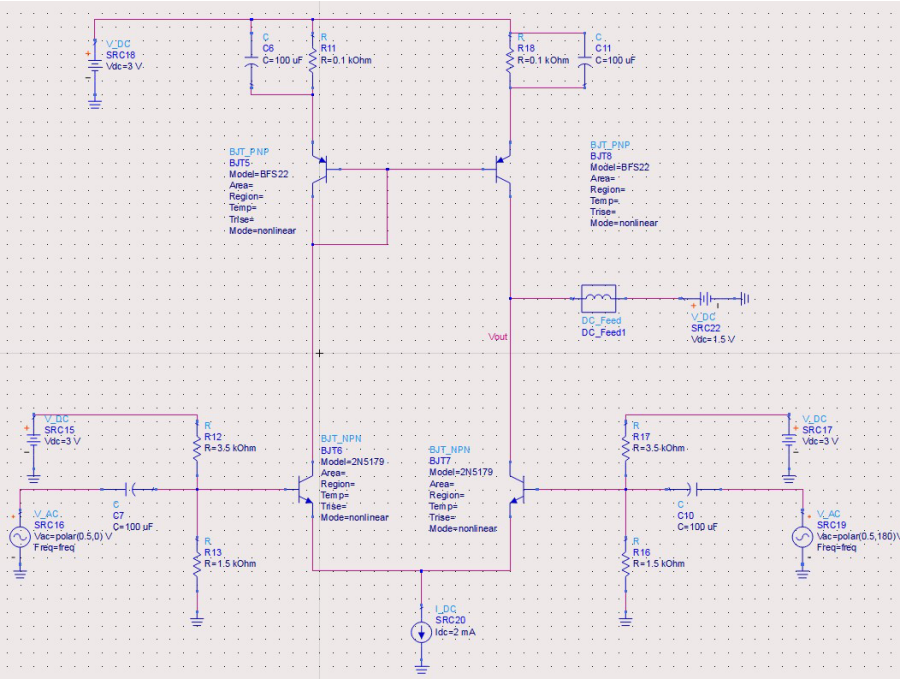


Figure 23: Differential amplifier with current mirror load

a. Determine the operating points of the devices and report the results. Additionally, compute the small-signal parameters and the values of parasitic capacitors using the given equations, based on the I_C values observed from the operating points.

$$C_\pi \approx g_m.TF + 2.C_{JE}$$

$$C_\mu \approx \frac{1}{r_o}.TR + C_{JC}$$

$$C_{j,BE} = \frac{C_{j,BE0}}{(1+|\frac{V_{BE}}{\phi_{i,BE}}|)^m}, \quad \text{with m} = \text{MJE}$$

$$C_{j,BC} = \frac{C_{j,BC0}}{(1+|\frac{V_{BC}}{\phi_{i,BC}}|)^m}, \quad \text{with m} = \text{MJC}$$

Like the previous problems, we sketch the desired circuit and do the DC simulation.

You can see the circuit and the resulting operating points in the figures in the next page.

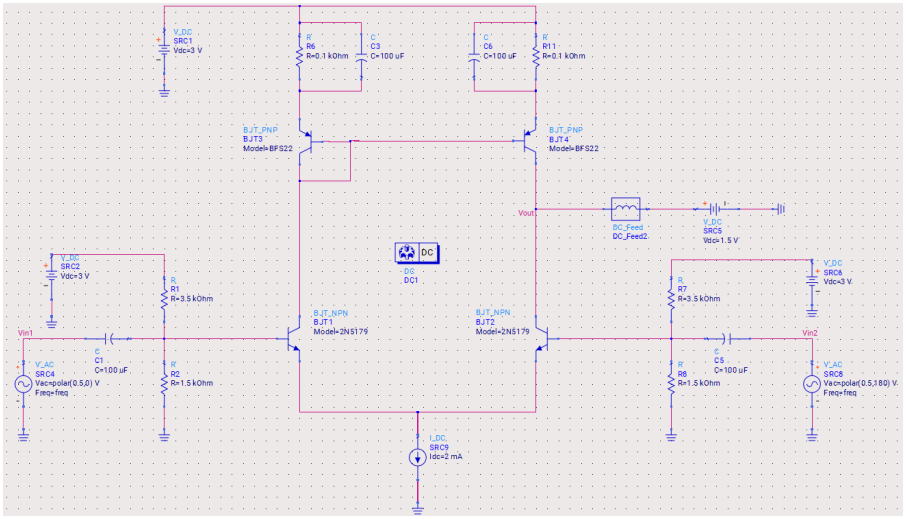


Figure 24: DC simulation circuit

index	Ic	Vbc	Vbe	Vce
1	989.5 u	-1.383	779.1 m	2.162
2	982.0 u	-615.0 m	779.1 m	1.394
3	-978.2 u	0.0000	-633.8 m	-633.8 m
4	-981.5 u	767.8 m	-633.5 m	-1.401

Figure 25: DC simulation results

	V_{CE}	I_C	g_m	r_π	r_o
Q_1	2.162^V	$989.5^{\mu A}$	$38.1^{m\mho}$	$7.41^{k\Omega}$	$101.1^{k\Omega}$
Q_2	1.394^V	$982^{\mu A}$	$37.8^{m\mho}$	$7.46^{k\Omega}$	$101.8^{k\Omega}$
Q_3	-0.634^V	$978.2^{\mu A}$	$37.6^{m\mho}$	$4.79^{k\Omega}$	$51.1^{k\Omega}$
Q_4	-1.401^V	$981.5^{\mu A}$	$37.8^{m\mho}$	$4.76^{k\Omega}$	$50.8^{k\Omega}$

Table 4: Simulation bias points and small-signal parameters

Now lets compute the values of the parasitic capacitors.

$$C_{\pi_1} = g_{m_1}.TF_1 + 2.\frac{C_{j,BE0}}{(1+\frac{V_{BE_1}}{\phi_{i,BE}})^m} = 38.1^{m\mho}.141^{psec} + 2.\frac{939.8^{fF}}{(1+\frac{779.1^{mV}}{0.75^V})^{0.3453}} \approx 6.84^{pF}$$

$$C_{\mu_1} \approx \frac{1}{r_{o_1}}.TR_1 + \frac{C_{j,BC0}}{(1+\frac{V_{BC_1}}{\phi_{i,BC}})^m} = \frac{1.588^{nsec}}{101.1^{k\Omega}} + \frac{893.1^{fF}}{(1+\frac{1383^{mV}}{0.75})^{0.3017}} \approx 0.667^{pF}$$

$$C_{\pi_2} = 37.8^{m\mho}.141^{psec} + 2.\frac{939.8^{fF}}{(1+\frac{779.1^{mV}}{0.75^V})^{0.3453}} \approx 6.8^{pF}, C_{\mu_2} \approx \frac{1.588^{nsec}}{101.8^{k\Omega}} + \frac{893.1^{fF}}{(1+\frac{615^{mV}}{0.75})^{0.3017}} \approx 0.761^{pF}$$

$$C_{\pi_3} = 37.6^{m\mho}.350^{psec} + 2.\frac{6^{pF}}{(1+\frac{633.8^{mV}}{0.7^V})^{0.4}} \approx 23^{pF}, C_{\mu_3} \approx \frac{10^{nsec}}{51.1^{k\Omega}} + \frac{4^{pF}}{(1+\frac{0^{mV}}{0.5})^{0.33}} \approx 3.15^{pF}$$

$$C_{\pi_4} = 37.8^{m\mho}.350^{psec} + 2.\frac{6^{pF}}{(1+\frac{633.5^{mV}}{0.7^V})^{0.4}} \approx 23^{pF}, C_{\mu_4} \approx \frac{10^{nsec}}{50.8^{k\Omega}} + \frac{4^{pF}}{(1+\frac{767.8^{mV}}{0.5})^{0.33}} \approx 3.15^{pF}$$

b. Derive the theoretical parametric transfer function of the circuit. Then, using the values obtained in the previous step, plot the **Bode diagram** of the circuit in *MATLAB*.

The figure below shows the AC circuit.(derived from the reference book)

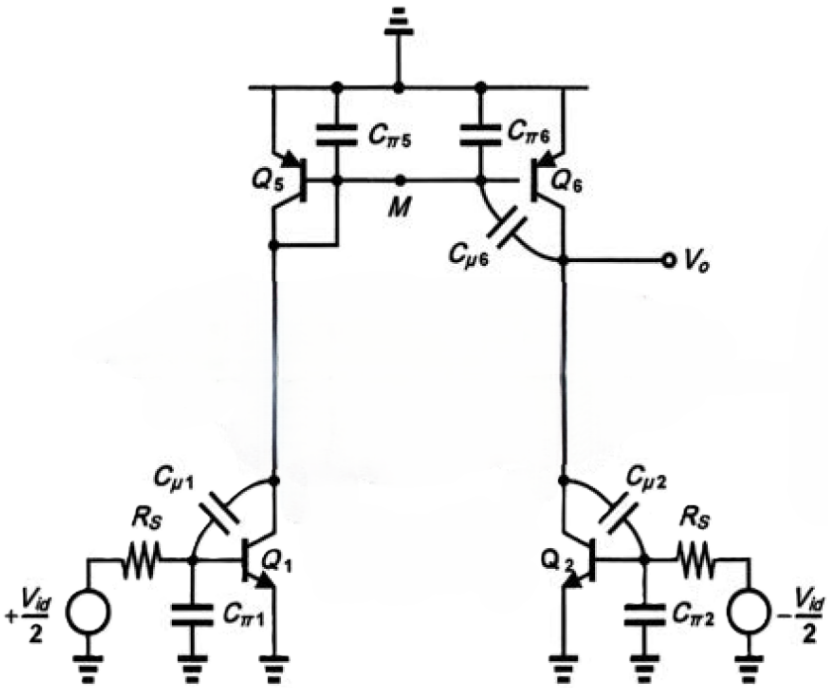


Figure 26: The AC circuit

We begin the process by extracting the poles associated with each node of the circuit. (In our problem $R_S = 0$ but there are two parallel resistors named R_1 and R_2 instead)

- Positive input node

  $P = \frac{1}{\tau} = \frac{1}{R_{eq}C_{eq}}$.

  $A_v = -g_{m_1}(r_{o_1}||r_{o_5}||\frac{1}{g_{m_5}}) \approx -1 \Rightarrow C_{eq} = C_{\pi_1} + (1 - A_v)C_{\mu_1} = C_{\pi_1} + 2C_{\mu_1}$
  $R_{eq} = R_1||R_2||r_{\pi_1} \approx 0.92^{k\Omega}, C_{eq} \approx 8.17^{pF} \Rightarrow P_{i+} = \frac{1}{8.17^{pF} \times 920^{\Omega}} \approx 133^{MHz}$

- Negative input node:

  $A_v = -g_{m_2}(r_{o_2}||r_{o_6}) \approx -1281 \Rightarrow C_{eq} = C_{\pi_2} + (1 - A_v)C_{\mu_2}$
  $C_{eq} \approx 982.4^{pF}, R_{eq} \approx 0.92^{k\Omega} \Rightarrow P_{i-} \approx 1.11^{MHz}$

- Output node

  This node is also common in the two ways from input to output and usually has the dominant pole of the circuit.

  $C_{eq} = 2(C_{\mu_2} + C_{\mu_6}) \approx 7.82^{pF}, R_{eq} = r_{o_2}||r_{o_6} \approx 33.89^{k\Omega} \Rightarrow P_o \approx 3.77^{MHz}$

- Current mirror node(M)

  The pole associated with this node is mostly much farther than the output pole. Therefor I assume the output node is connected to ground.

  $C_{eq} = C_{\pi_5} + C_{\pi_6} + 2C_{\mu_6} + 2C_{\mu_1} \approx 50.48^{pF}, R_{eq} = r_{\pi_6}||r_{o_1}||\frac{1}{g_{m_5}} \approx 26.4^{\Omega}$
  $\Rightarrow P_M \approx 749^{MHz}$

* Zeros

$$Z_+ = \frac{g_{m_1}}{C_{\mu_1}} \approx 57.1^{GHz} \quad , \quad Z_- = \frac{g_{m_2}}{C_{\mu_2}} \approx 49.7^{GHz}$$

So the general form of the transfer function would be like:

$$A_v(s) = \frac{V_{out}}{V_{id}} = A_{v_0}\frac{V_{o1}+V_{o2}}{V_{id}} = A_{v_0}\frac{(\frac{1-\frac{s}{Z_+}}{(1+\frac{s}{P_{i+}})(1+\frac{s}{P_M})(1+\frac{s}{P_o})})\frac{V_{id}}{2} - (\frac{1-\frac{s}{Z_-}}{(1+\frac{s}{P_{i-}})(1+\frac{s}{P_o})})\frac{-V_{id}}{2}}{V_{id}} \Rightarrow$$

$$A_v(s) = \frac{A_{v_0}}{2}.\frac{(1-\frac{s}{Z_+})(1+\frac{s}{P_{i-}})+(1-\frac{s}{Z_-})(1+\frac{s}{P_{i+}})(1+\frac{s}{P_M})}{(1+\frac{s}{P_{i+}})(1+\frac{s}{P_{i-}})(1+\frac{s}{P_o})(1+\frac{s}{P_M})} \text{ where: } A_{v_0} = G_M R_{out} \approx 1281$$

Now we put in the numerical values and plot the Bode diagram of the derived transfer function in *MATLAB*.

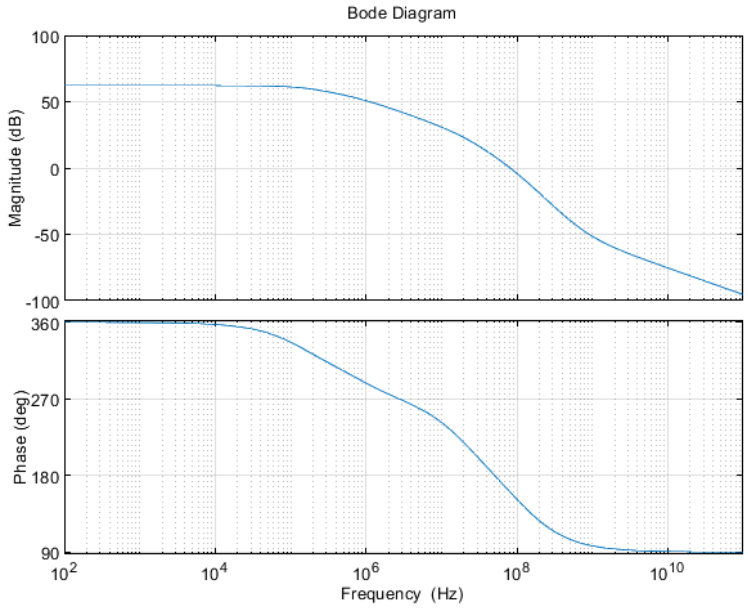


Figure 27: Differential-mode frequency response of the derived transfer function in *MATLAB*

c. Describe the **Mirror pole** and its effects in this circuit.

P_M appears only in one of the ways to the output node. This creates an extra zero in the transfer function and here we have 3 zeros and 4 poles in contrast with the circuit in the previous problem.

d. Perform an AC simulation in *ADS* and plot the **differential frequency response** of the circuit.

The corresponding circuit and the results:

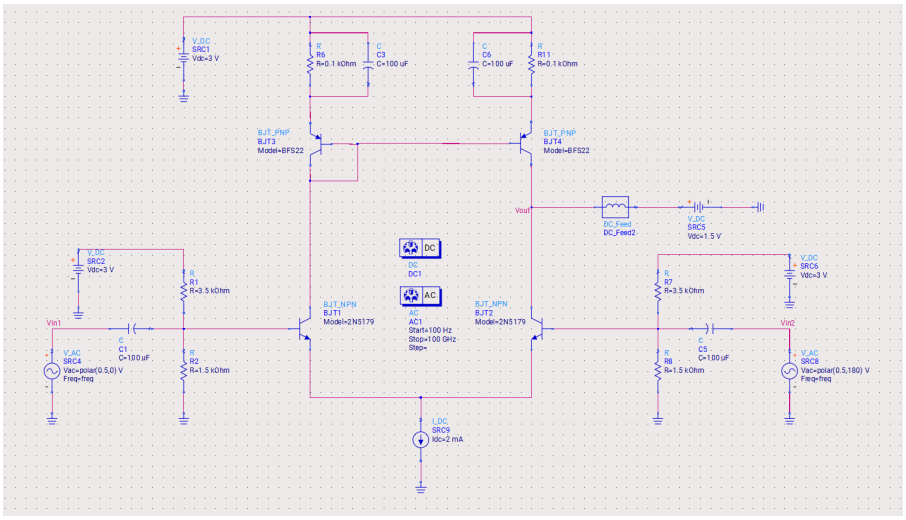


Figure 28: AC simulation circuit

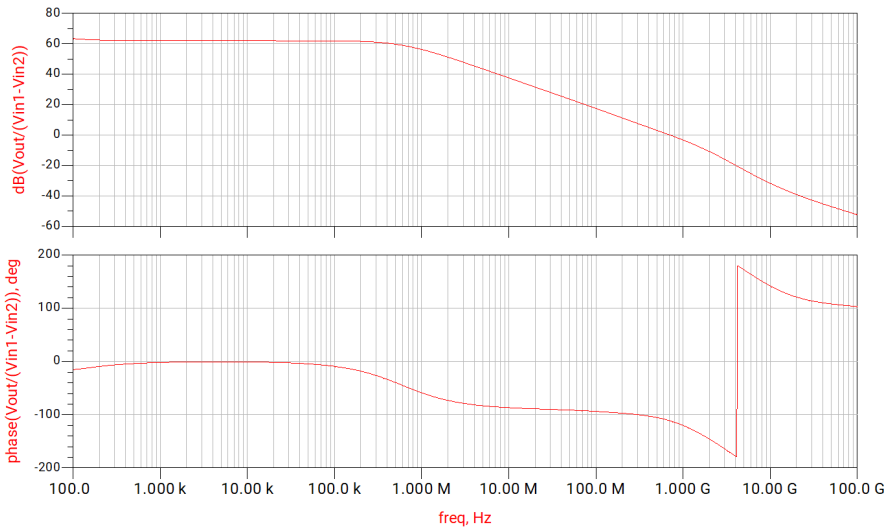


Figure 29: Differential-mode frequency response of the circuit in *ADS*

e. Compare the transfer function obtained from *MATLAB* in part *b* with the frequency response simulated in *ADS* in part *d* and explain the differences between the theoretical and simulated results.

We see from the results that the general form of both Bode diagrams are similar. There are some differences in Bandwidth and also high frequency gain that have arisen from the fact that we didn't consider the By-Pass capacitors and also from the Miller Approximations we used to associate each pole with a certain node.

f. Plot the **common-mode frequency response** in *ADS* and compare it to the results from the question 2.

The resulting Bode diagram is visible on figure 30

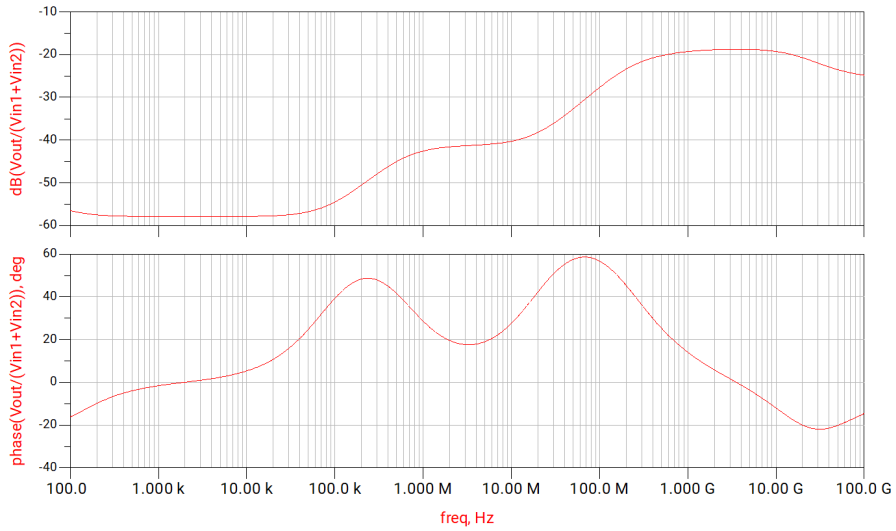


Figure 30: Common-mode frequency response of the circuit in *ADS*

And that of problem 2:(both with ideal current source)

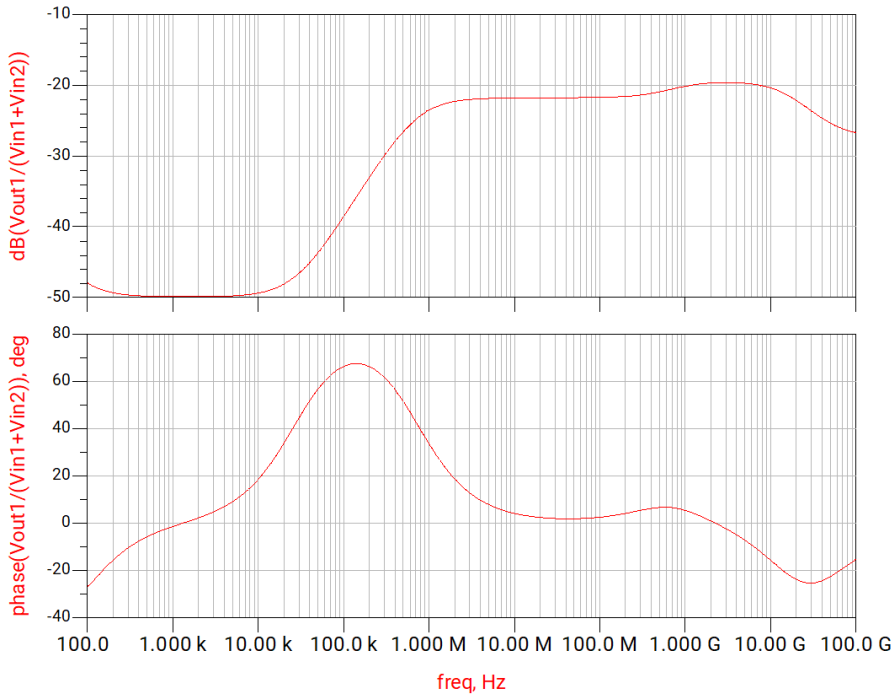


Figure 31: Common-mode frequency response of the circuit in problem 2 in *ADS*

As we can see, the common-mode gain is lower than that of problem 2 and remains low for higher frequencies which means a better *CMRR*.

In both diagrams we have a zero in low frequencies which is why we initially see an increase in the gain.

Problem 4

In this problem, you will design an amplifier using the 180nm *CMOS* process and explore its performance through various simulations.

Consider the following schematic, which represents a differential amplifier using 180nm *CMOS* technology. Your task is to determine the transistors' sizes, resistors' values, I_{DC} amplitude, and input DC voltage while assuming $V_{DD} = 1.8^{V}$, in order to meet the following specifications:

- **Low-Frequency Differential Mode Gain(double-ended):** $>$ 20dB
- **Power Consumption:** $<$ 3.6 mW

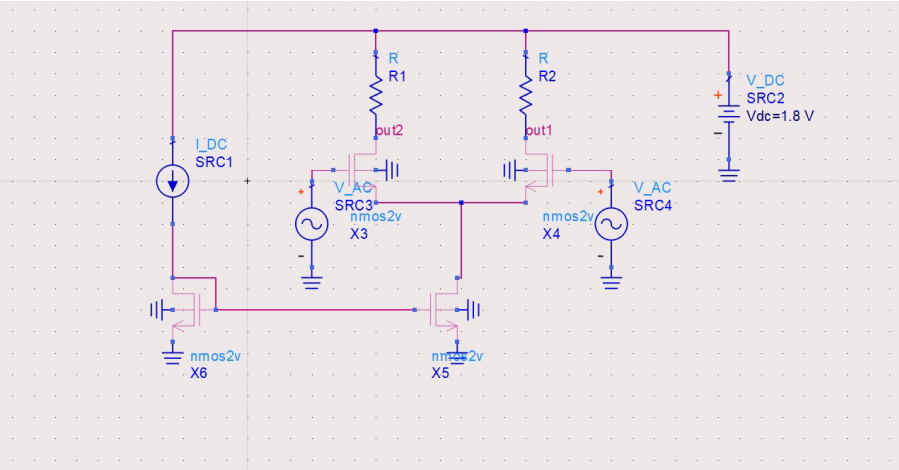


Figure 32: Differential amplifier Schematic

a. Write the **theoretical calculations** required for designing this circuit.

First of all, lets see what the constraints give us:
$P < 3.6^{mW} \Rightarrow 1.8^{V} \times (2I_D + I_{SS}) < 3.6^{mW} \Rightarrow 2I_D + I_{SS} < 2^{mA}$
$A_v > 20^{dB} = 10 \Rightarrow g_m(r_o||R_D) > 10 \Rightarrow \frac{2I_D}{V_{od}}(\frac{1}{\lambda I_D}||R_D) > 10$

Lets begin the designing process now.

I like to choose $I_{SS} = 0.4^{mA}$ so that I_D could be up to 0.8^{mA} so that we have a higher g_m for more gain. $I_{D_4} = 0.4^{mA}$ and $I_{D_3} < 1.6^{mA}$ so $\frac{(W/L)_3}{(W/L)_4} < 4$ because $V_{gs_3} = V_{gs_4}$. V_{DS_3} must be greater than V_{od_3} so I set $V_{od_4} = V_{od_3} = 0.225^{V} \rightarrow (W/L)_4 = \frac{11.1^{\mu m}}{180^{nm}}$. To make sure that the power does not exceed the limit while dealing with the gain I set $(W/L)_3 = (W/L)_4 \times 3.6 = \frac{40^{\mu m}}{180^{nm}}$. Assuming $V_{th} = 0.6^{V}$ we would have $I_{D_3} \approx 1.4^{mA} \Rightarrow P \approx 3.2^{mW}$. Everything is good so far. Now we shall deal with the circuit's gain.
For M_3 to work properly V_{D_3} must be greater than 0.225^{V}. So we need to give the inputs some DC voltage in order to make sure all the transistors work properly and are in the saturation region. I set this DC input level to

1^V. V_{D_3} is around 0.3^V to 0.4^V so $V_{gs_1} = 1 - 0.35 = 0.65 \Rightarrow V_{od_1} \approx 0.05^V$. I_{D_3} is 1.4^{mA} so to have $I_{D_1} \approx 0.7^{mA}$, $(W/L)_1$ needs to be $\frac{100^{\mu m}}{180^{nm}}$. This way $g_{m_1} = \frac{0.7}{2(0.05)} \approx 7^{m\mho}$. Thus we need to have $R_D > 1.4^{k\Omega}$ and also $R_D < \frac{1.8^V - (1^V - 0.65^V + 0.05^V)}{0.7^{mA}} = 2^{k\Omega}$. I chose $R_D = 1.66^{k\Omega}$ to make sure the devices remain in the saturation region. Using these values we get $A_v \approx 11.6$ which fulfills our desire.

The designed circuit looks like this:

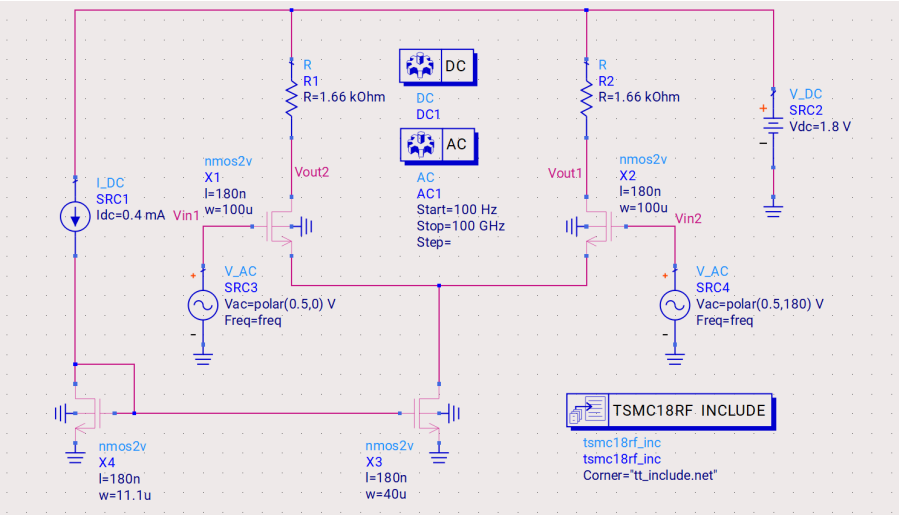


Figure 33: The designed circuit of problem 4

b. Verify the power consumption condition by analyzing the operating points of the devices.

index	Id	Vds	Vgs - Vth	Vgs	Vth	index	....Power
1	656.0 u	335.3 m	11.90 m	624.2 m	612.3 m	1	-3.082 m
2	656.0 u	335.3 m	11.90 m	624.2 m	612.3 m	2	0.0000
3	1.312 m	375.8 m	139.8 m	655.5 m	515.7 m	3	0.0000
4	400.0 u	655.5 m	140.7 m	655.5 m	514.9 m		

Figure 34: DC simulation results and total power

According to the results from DC simulation in the figure above, all the devices are active and in saturation. I_{D_3} has a 0.08^{mA} error which is due to our approximate calculations. The power is a good 3.082^{mW}.

	Theory			Simulation		
	V_{od}	I_D	g_m	V_{od}	I_D	g_m
M_1	0.05^V	$700^{\mu A}$	$7^{m\mho}$	0.012^V	$656^{\mu A}$	$10.8^{m\mho}$
M_2	0.05^V	$700^{\mu A}$	$7^{m\mho}$	0.012^V	$656^{\mu A}$	$10.8^{m\mho}$
M_3	0.225^V	1.4^{mA}	-	0.140^V	1.31^{mA}	-
M_4	0.225^V	$400^{\mu A}$	-	0.141^V	$400^{\mu A}$	-

Table 5: Comparison between theoretical values and simulation results

c. Plot the differential mode gain in *ADS* and adjust the circuit to achieve the desired gain.

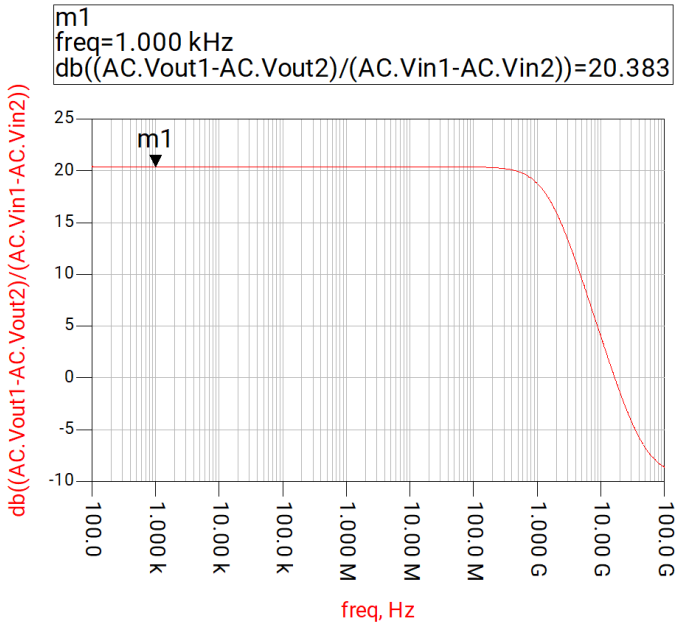


Figure 35

Despite the differences in the bias points and currents, the gain is still above 20^{dB} and the power is also good and therefore no adjustment is needed.

$$Gain = 20.38^{dB} \qquad Power = 3.08^{mW}$$

d. Determine f_{3dB} and approximate the poles and zeros of the circuit.

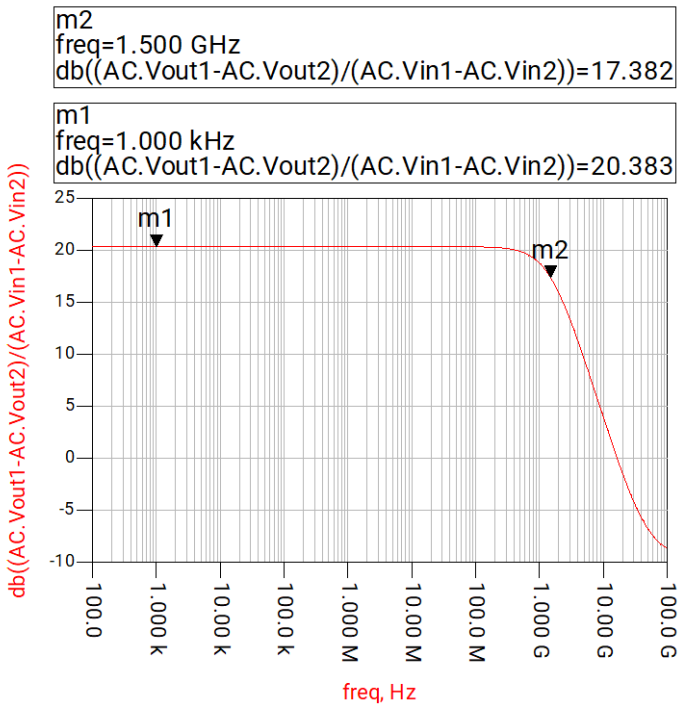


Figure 36

As reported by the figure above $f_{3dB} = 1.5^{GHz}$.

e. Plot the differential mode gain and common mode gain simultaneously and analyze the frequency behavior of the Common-Mode Rejection Ratio (*CMRR*).

I used the history option in *ADS* in order to plot both Differential-mode and common-mode frequency responses. The results are shown in the following figure.

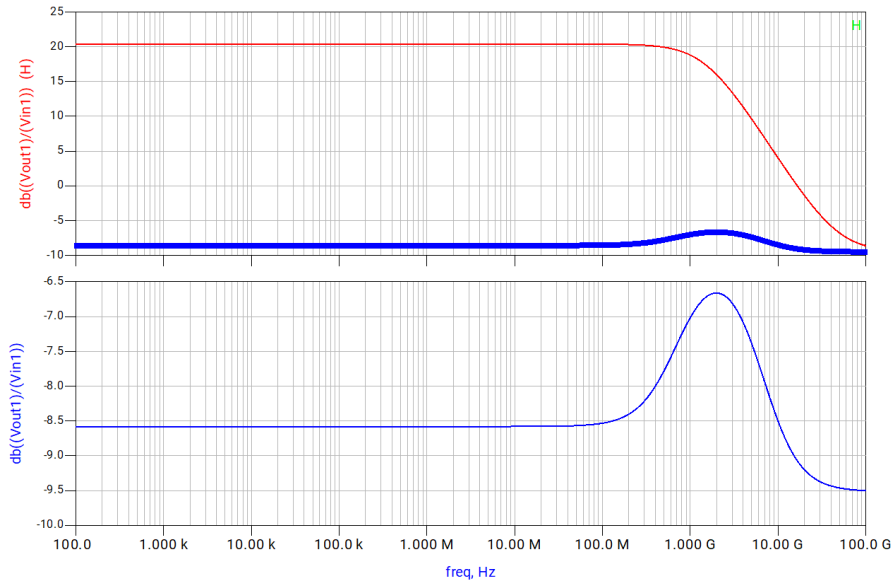


Figure 37: Bode diagram of differential-mode and common-mode simultaneously

As we can see the circuits has a good *CMRR* in a pretty frequency range which is great.

f. Compute the *CMR* of the circuit using *ADS*.

I swept the input DC voltage from 0.1^V to 1.8^V and observed the resulting currents for each transistor. We also have to make sure the gain does not change dramatically. The results are shown below.

index	Id
Vx=100.0 m, freq=0.0000 Hz	
1	32.91 n
2	32.91 n
3	65.74 n
4	400.0 u
Vx=200.0 m, freq=0.0000 Hz	
1	568.6 n
2	568.6 n
3	1.136 u
4	400.0 u
Vx=300.0 m, freq=0.0000 Hz	
1	8.490 u
2	8.490 u
3	16.96 u
4	400.0 u
Vx=400.0 m, freq=0.0000 Hz	
1	73.26 u
2	73.26 u
3	146.4 u
4	400.0 u
Vx=500.0 m, freq=0.0000 Hz	
1	250.0 u
2	250.0 u
3	499.9 u
4	400.0 u
Vx=600.0 m, freq=0.0000 Hz	
1	438.9 u
2	438.9 u
3	877.8 u
4	400.0 u

index	Id
Vx=700.0 m, freq=0.0000 Hz	
1	545.7 u
2	545.7 u
3	1.091 m
4	400.0 u
Vx=800.0 m, freq=0.0000 Hz	
1	597.8 u
2	597.8 u
3	1.196 m
4	400.0 u
Vx=900.0 m, freq=0.0000 Hz	
1	631.0 u
2	631.0 u
3	1.262 m
4	400.0 u
Vx=1000.0 m, freq=0.0000 Hz	
1	656.0 u
2	656.0 u
3	1.312 m
4	400.0 u
Vx=1.100 , freq=0.0000 Hz	
1	676.6 u
2	676.6 u
3	1.353 m
4	400.0 u
Vx=1.200 , freq=0.0000 Hz	
1	694.2 u
2	694.2 u
3	1.388 m
4	400.0 u

index	Id
Vx=1.300 , freq=0.0000 Hz	
1	706.3 u
2	706.3 u
3	1.413 m
4	400.0 u
Vx=1.400 , freq=0.0000 Hz	
1	710.2 u
2	710.2 u
3	1.420 m
4	400.0 u
Vx=1.500 , freq=0.0000 Hz	
1	711.5 u
2	711.5 u
3	1.423 m
4	400.0 u
Vx=1.600 , freq=0.0000 Hz	
1	712.0 u
2	712.0 u
3	1.424 m
4	400.0 u
Vx=1.700 , freq=0.0000 Hz	
1	712.3 u
2	712.3 u
3	1.425 m
4	400.0 u
Vx=1.800 , freq=0.0000 Hz	
1	712.4 u
2	712.4 u
3	1.425 m
4	400.0 u

Figure 38: The resulting currents by sweeping the input DC level from 0.1^V to 1.8^V

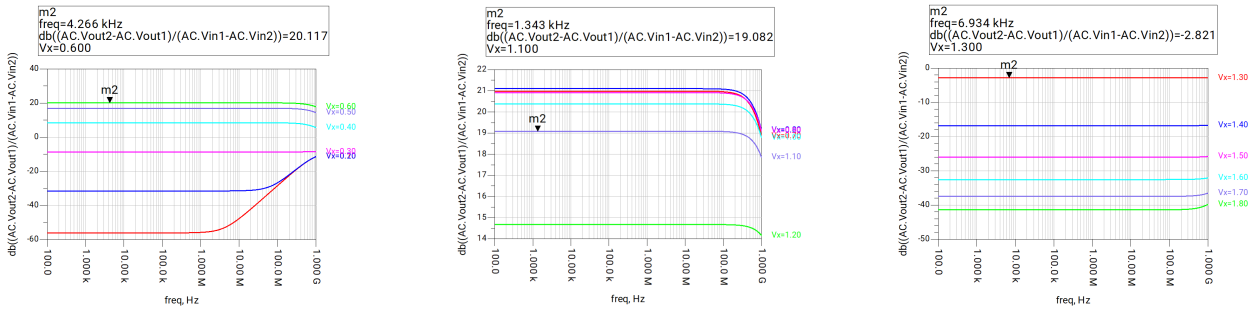


Figure 39: The resulting gains by sweeping the input DC level from 0.1^V to 1.8^V

It is observed that for $V_{CM} = [1^V - 1.8^V]$ the currents remain nearly the same while the gain remains within $1dB$ from the base 20^{dB} for $V_{CM} = [0.6^V - 1.1^V]$.

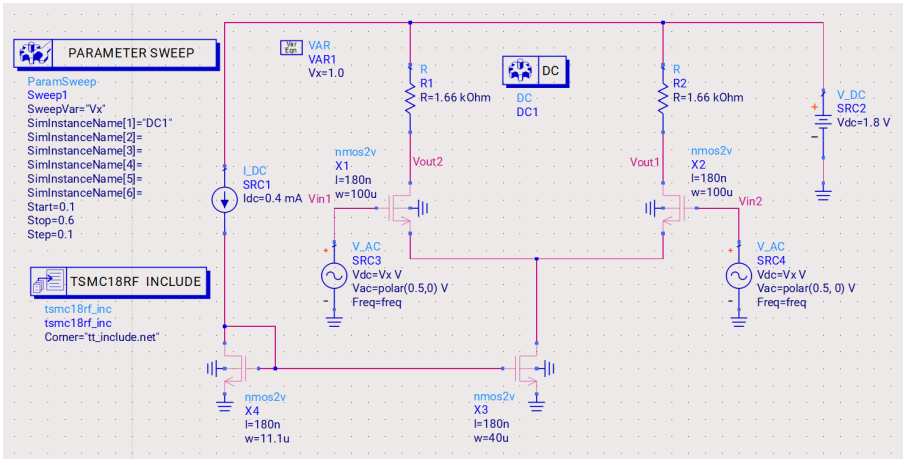


Figure 40: CMR simulation circuit

g. Using transient simulation, determine the symmetric output swing.

In this part I used time-domain sources and transient analysis to measure the circuit's swing.

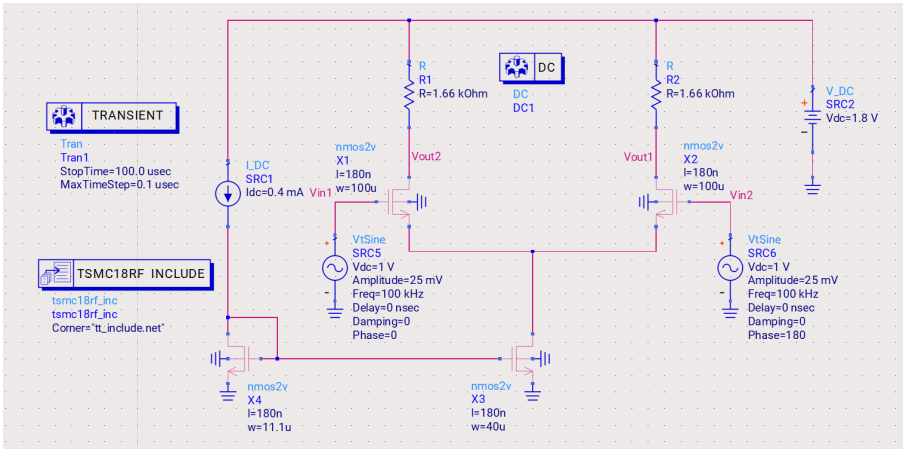


Figure 41: Transient simulation circuit

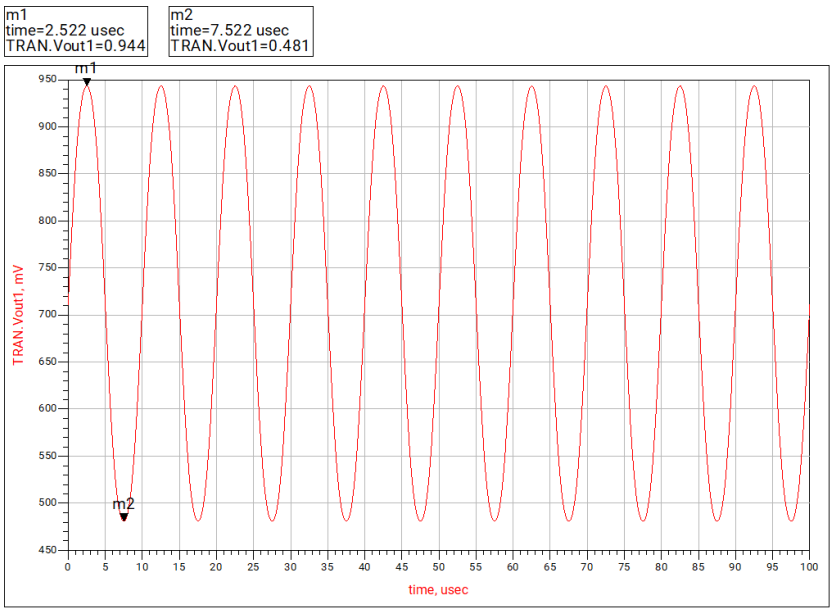


Figure 42: Transient simulation results

To see the maximum symmetric swing I increased the input voltage magnitude gradually until when I saw the output starting to get nonlinear.

It is obvious from the results that the output symmetric swing is:

$[S^+, S^-] = [0.481^V, 0.944^V] \Longrightarrow Swing = 232^{mV}$

which isn't so good and shows that this design still needs to be adjusted to get a higher swing.